Supplementary Material:

iSpa3D: An interpretable deep learning framework for 3D spatial domain reconstruction and domain-specific gene discovery in multi-slice spatial transcriptomics

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1 Supplementary Text

S1 Dataset

To evaluate the effectiveness and generalization capability of iSpa3D, we applied it to multiple datasets generated by different spatial transcriptomics (ST) platforms (Supplementary Table S1) [1].

Spatial transcriptomics sequencing of **the human dorsolateral prefrontal cortex (DLPFC)** was performed using the 10x Genomics Visium platform [2]. Dataset was derived from three adult donors with normal neurological phenotypes. Four adjacent sections were collected from each donor (i.e., each pair of sections represented sequentially adjacent slices with slight spatial offset between pairs), totaling 12 sections for analysis. The number of probe spots per section ranged from 3,498 to 4,789. In the original paper, Maynard et al [3]. manually annotated each probe spot according to cortical layer (L1–L6) and white matter (WM) status, treating this annotation as the ground truth for layer identification.

The Mouse embryo dataset (MOSTA project), generated by Stereo-seq, encompasses eight developmental stages (E9.5 to E16.5) at near-single-cell resolution. To capture the critical dynamics of early development and ensure compatibility with iSpa3D model, our analysis focused on the first three stages (E9.5 to E11.5) [4]. Within this subset, the number of spots increased substantially from approximately 5,913 to 25,201, while the gene detection coverage remained consistently around 25,000.

The mouse whole-brain spatial transcriptomics dataset was generated using the ST platform for spatial transcriptomics sequencing [5]. It comprises 35 coronal sections aligned along the anteroposterior (AP) axis, covering the entire brain structure from the olfactory bulb to the cerebellar bud. Ortiz et al. provided manually annotated clusters across 15 categories within this dataset, serving as a reference standard for tissue architecture and cell types [6].

The mouse brain sagittal section dataset for section 1 was sequenced using the 10x Visium platform for spatial transcriptomics, covering anterior and posterior sections comprising approximately 2,695 and 3,353 spots, respectively. The anterior section provides manually annotated ground truth for evaluating and validating spatial transcriptomics analysis methods [7].

The mouse hypothalamic anterior region dataset was sequenced using MERFISH technology for spatial transcriptomics [8], covering five consecutive tissue sections located at Bregma coordinates -0.04, -0.09, -0.14, -0.19, and -0.24 mm. Each section contained expression data for approximately 155 genes, with 5,488 to 5,926 spots. Manual annotations were provided for each section, delineating eight major brain regions [9].

The mouse main olfactory bulb (MOB) [4] was profiled in two publicly available spatial transcriptomics datasets generated on the Stereo-seq and Slide-seqV2 [10] platforms, which were analyzed in their preprocessed versions released in the SEDR [7] study. The Stereo-seq dataset contains 19,109 spatial spots and 27,106 genes, whereas the Slide-seqV2 dataset comprises 21,724 spatial spots and 21,220 genes. Biological prior information on MOB laminar organization was obtained from DAPI-stained coro-

nal sections and the anatomical annotations of the Allen Mouse Brain Atlas, which served as reference for spatial domain identification and cross-platform alignment. For layer-level annotation, we followed the canonical seven-layer organization of the mouse MOB in coronal sections: olfactory nerve layer (ONL), glomerular layer (GL), external plexiform layer (EPL), mitral cell layer (MCL), internal plexiform layer (IPL), granule cell layer (GCL), and the rostral migratory stream (RMS).

The mouse Alzheimer's disease (AD) spatial transcriptomics dataset was generated using the STARmap PLUS platform [11], an imaging-based technology capable of mapping gene expression at single-cell resolution. The dataset captures the spatiotemporal dynamics of neurodegeneration by comprising brain sections from the TauPS2APP mouse model and wild-type controls across two distinct disease stages: 8 months and 13 months. Our analysis utilized four representative sections (covering 8-month Control/AD and 13-month Control/AD conditions) to investigate disease-induced structural remodeling. These sections provide high-resolution spatial profiles for thousands of single cells. Detailed anatomical annotations based on histological landmarks were used as ground truth, delineating key brain structures including the isocortex layers, corpus callosum (cc), and fine-grained hippocampal subregions (CA1, CA2, CA3, and Dentate Gyrus), serving as the reference standard for evaluating cross-condition alignment and 3D spatial domain reconstruction.

The Axolotl telencephalon spatial transcriptomics dataset was generated using the Stereo-seq platform [12], achieving single-cell resolution with a large field of view. This comprehensive atlas maps the cellular dynamics of the axolotl forebrain across six developmental stages (from Stage 44 to Adult) and seven injury-induced regenerative stages. To evaluate the capability of iSpa3D in capturing long-term developmental transitions and structural maturation, our analysis focused on the critical comparison between the Juvenile and Adult stages. These selected sections contain approximately 11,698 and 8,243 single cells, respectively, with high gene detection sensitivity. In the original study, Wei et al. provided detailed annotations of 33 distinct cell types during development, including subtypes of excitatory neurons and ependymoglial cells.

S2 Data preprocessing

Gene expression matrices and spatial coordinates were organized by section and batch. Spots without valid reference annotations were removed before training. Genes shared across the sections to be integrated were retained, and the section-specific matrices were concatenated using Scanpy [13].

In the DLPFC workflow, raw counts were preserved in a separate data layer. Genes detected in fewer than 50 spots or with fewer than 10 total counts were removed. Expression values were normalized per spot to a total of 10^6, while highly variable genes were selected from the preserved raw-count layer using the Seurat v3 strategy. The selected normalized expression features were standardized by z-scoring without an intervening log transformation. PCA was performed with 200 components and a random

state of 42 [14]. The number of highly variable genes was specified in the experiment configuration.

For MERFISH, the measured gene features were used directly without PCA. Thus, $\mathbf{X} \in \mathbb{R}^{N\times m}$ denotes the model input, with m representing either the number of retained principal components or the number of input genes. PCA loadings were retained for gene-space projection of attribution scores only when PCA was applied.

S3 Spatial graph construction and masked representation learning

Section-wise spatial graphs. For section s, $\mathbf{P}^{(s)} \in \mathbb{R}^{N_s\times 2}$ contains the spot coordinates. Let $\mathcal{G}^{(s)} = (\mathcal{V}^{(s)}, E^{(s)})$ denote its spatial graph and $E = \bigcup_s E^{(s)}$ the combined edge set. The adjacency matrix $\mathbf{A}$ is assembled from E, and $\tilde{\mathbf{A}} = \mathbf{D}^{-1/2}(\mathbf{A}+\mathbf{I})\mathbf{D}^{-1/2}$, where $\mathbf{D}$ is the degree matrix of $\mathbf{A}+\mathbf{I}$. Spatial graphs are constructed independently for each section using the Euclidean distances between spot coordinates. Each spot is connected to its k nearest neighbors, and the resulting graph is symmetrized. The section-specific graphs are combined into a disconnected multi-section graph, with no spatial edges between different sections. This construction preserves within-section neighborhoods without requiring prior spatial registration. Cross-section relationships are introduced through the representation-learning objectives rather than through the initial spatial graph.

Complementary feature masking. Let $\mathcal{V}$ denote the spot index set and ρ the masking rate. A random permutation partitions the spots into a subset $\mathcal{V}_m$ of size $\lfloor \rho N \rfloor$ and its complement $\mathcal{V}_{cm}$, satisfying

$$\mathcal{V}_m \cup \mathcal{V}_{cm} = \mathcal{V}, \qquad \mathcal{V}_m \cap \mathcal{V}_{cm} = \emptyset. \tag{S1}$$

Two views are constructed by corrupting features in $\mathcal{V}_m$ and $\mathcal{V}_{cm}$, respectively, while retaining the original features outside each subset. In the complementary-view implementation, approximately 90% of the selected spots receive a learnable mask token. The remaining 10% receive features sampled from other spots, with the mask token added to these replacement features. The spatial graph is unchanged. The two views are passed through the same encoder and cluster head for CLCL.

The reconstruction branch independently samples a masked spot set $\mathcal{V}_{\text{rec}}$ using the same masking rate and constructs a corrupted input $\tilde{\mathbf{X}}_{\text{rec}}$. In this branch, token-selected spots receive the learnable token, whereas replacement-selected spots receive randomly sampled spot features. Reconstruction loss is evaluated only at the selected spots. The complementary-view and reconstruction branches therefore share model parameters but need not use the same sampled masks in a training iteration.

Graph encoder and decoder. The encoder consists of two feed-forward blocks followed by two graph convolutional layers. For $v \in \{m, cm\}$, the encoder and decoder are written as $\mathbf{Z}_v = F_g(\tilde{\mathbf{X}}^{(v)}, \tilde{\mathbf{A}})$ and $\hat{\mathbf{X}}_{\text{rec}} = G_d(\tilde{\mathbf{A}}, \mathbf{Z}_{\text{rec}})$, respectively, where $\mathbf{Z}_{\text{rec}} = F_g(\tilde{\mathbf{X}}_{\text{rec}}, \tilde{\mathbf{A}})$ and the projection is included in G_d.

Let $\mathbf{H}_f^{(0)} = \tilde{\mathbf{X}}^{(v)}$. The feed-forward blocks are

$$\mathbf{H}_f^{(l)} = \mathrm{Dropout}\Big(\mathrm{ELU}\Big(\mathrm{BN}\Big(\mathbf{H}_f^{(l-1)}\mathbf{W}_f^{(l)} + \mathbf{b}_f^{(l)}\Big)\Big)\Big), \qquad l = 1, 2, \tag{S2}$$

where BN denotes batch normalization. Weight matrices are written for row-wise spot representations, and biases are broadcast across spots. The graph convolutional layers then aggregate spatial information:

$$\begin{aligned} \mathbf{H}_g &= \mathrm{Dropout}\Big(\mathrm{ReLU}\Big(\mathrm{BN}\Big(\mathrm{GCN}_1(\mathbf{H}_f^{(2)}, E)\Big)\Big)\Big), \\ \mathbf{Z}_v &= \mathrm{Dropout}(\mathrm{BN}(\mathrm{GCN}_2(\mathbf{H}_g, E))), \end{aligned} \tag{S3}$$

where E is the combined spatial edge set. Each GCN layer performs neighborhood aggregation with self-loops and symmetric degree normalization. The first graph layer uses ReLU activation, whereas the second has an identity activation. Dropout is applied during training.

Before decoding, a bias-free linear projection maps the latent embeddings to the decoder input. A single graph convolutional layer reconstructs the input features:

$$\hat{\mathbf{X}}_{\mathrm{rec}} = \mathrm{Dropout}(\mathrm{BN}(\mathrm{GCN}_{\mathrm{dec}}(\mathbf{Z}_{\mathrm{rec}}\mathbf{W}_{\mathrm{proj}}, E))), \tag{S4}$$

where the decoder uses an identity activation and returns the same feature dimension as the model input.

The shared cluster head comprises a bias-free linear layer, batch normalization, GELU activation, and a second linear layer with bias. Its logits are converted to cluster probabilities by temperature-scaled softmax. The cluster-assignment and prototype contrastive objectives are given in the main Methods. The contrastive denominator includes prototypes from both views, excluding the anchor itself; the matching cluster in the complementary view is the positive. The marginal regularizer is $K^{-1}\sum_{v\in\{m,cm\}}\mathrm{KL}(\mathbf{u}\|\bar{\mathbf{p}}_v)$.

S4 Cross-section correspondence and spatial registration

Cross-section triplet construction. During training, the current embeddings are clustered to obtain provisional domain labels. For each section, candidate matches are identified among spots in all other sections by ranking latent dot products. Candidate pairs are randomly subsampled and retained only when their provisional domain labels agree. Duplicate pairs are removed, and the retained pairs define an undirected correspondence graph. For each anchor, one connected spot is selected as the positive, while a negative is sampled from the anchor's own section. These triplets are refreshed periodically after warm-up to reflect changes in the learned representations. The triplet loss is defined in the main Methods.

Latent-guided spatial correspondences. After representation learning, sections are registered progressively in the specified section order. For each pair of consecutive sections in this order, spots are

sampled without replacement using a common sample size $n_s = \min(5000, \min_t N_t)$, where N_t is the number of spots in section t. The registration routine uses a default random seed of 1234. Approximate nearest-neighbor searches are performed in the latent space using Annoy with Euclidean distance and one neighbor per query. Pairs are retained when the nearest-neighbor relationship is reciprocal.

The resulting pairs provide correspondences between the original spatial coordinates of the moving section and the current registered coordinates of the reference section. Thus, latent features determine which spots are matched, while the rigid transformation is estimated in the two-dimensional spatial coordinate system. If no reciprocal pairs are found, that pairwise alignment is skipped.

Rigid transformation and progressive assembly. Let $\mathbf{a}_\ell, \mathbf{b}_\ell \in \mathbb{R}^2$ denote the moving and reference coordinates of matched pair ℓ, respectively. A rotation $\mathbf{R}$ and translation $\mathbf{t}$ are estimated by minimizing

$$\min_{\mathbf{R},\mathbf{t}} \sum_{\ell=1}^{L} \|\mathbf{R}\mathbf{a}_\ell + \mathbf{t} - \mathbf{b}_\ell\|_2^2, \qquad \mathbf{R}^\top\mathbf{R} = \mathbf{I}, \quad \det(\mathbf{R}) = 1, \tag{S5}$$

where L is the number of matched pairs. After subtracting the coordinate centroids $\bar{\mathbf{a}}$ and $\bar{\mathbf{b}}$, the cross-covariance matrix and its singular value decomposition are

$$\mathbf{H} = \sum_{\ell=1}^{L} (\mathbf{a}_\ell - \bar{\mathbf{a}})(\mathbf{b}_\ell - \bar{\mathbf{b}})^\top = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top. \tag{S6}$$

The rigid solution is

$$\mathbf{R} = \mathbf{V}\,\mathrm{diag}\big(1, \det(\mathbf{V}\mathbf{U}^\top)\big)\,\mathbf{U}^\top, \qquad \mathbf{t} = \bar{\mathbf{b}} - \mathbf{R}\bar{\mathbf{a}}, \tag{S7}$$

with the determinant correction preventing reflection. The transformation is applied to all spots in the moving section. The first section defines the reference frame, and each subsequent section is aligned to the already registered preceding section.

For 3D visualization, registered two-dimensional coordinates are combined with a section-level z coordinate. Physical inter-section spacing is used when available; otherwise, section order defines a comparative stack. Domain labels and gene expression values are retained for visualization in the registered coordinate frame.

S5 Domain-specific gene attribution and selection

Domain prediction and neighborhood baselines. A domain classifier maps encoder representations to logits for the inferred spatial domains. Attribution targets the pre-softmax logit $S_c(\mathbf{x}_i)$ of domain c for spot i and is evaluated with the encoder and classifier in evaluation mode, without feature masking. Reference inputs are constructed from neighboring domains to relate feature contributions to the surrounding tissue context.

Neighboring domains are identified separately within each section using a spatial radius R. For target

domain c and candidate neighbor u, the procedure counts spot pairs whose spatial distance is at most R. A candidate qualifies within a section when this count exceeds a specified ratio τ_{nbr} of the total number of spots in u. Because the count is over pairs, a neighboring spot can contribute more than once. Domains qualifying in more than half of the sections containing c are retained. If none passes this majority criterion but candidates exist, the domain receiving the most section-level votes is selected.

The mean input feature vector of each retained domain provides a baseline:

$$\mathcal{B}_c = \left\{ \bar{\mathbf{x}}_u : \bar{\mathbf{x}}_u = \frac{1}{|\mathcal{V}_u|} \sum_{i \in \mathcal{V}_u} \mathbf{x}_i, \quad u \in \text{NBR}(c) \right\}, \tag{S8}$$

where $\mathcal{V}_u$ is the spot set of domain u. The implementation limits the number of baselines to three by default, selecting evenly spaced entries from the neighbor list ordered by vote count. Domains without a valid baseline are skipped.

Attribution methods. Integrated Gradients (IG) [15], Occlusion (OC) [16], and GradientShap (GS) [17] are applied to the input features through the encoder and domain classifier. IG integrates input gradients along a straight-line path from a baseline to the observed input, using 50 integration steps. When several baselines are available, the resulting IG attributions are averaged across baselines.

OC replaces one input feature at a time with its baseline value and measures the change in the target-domain logit. The mean of the available baseline vectors is used for occlusion, with a feature-window size and stride of one. GS samples reference inputs from the baseline set and interpolation points between the reference and observed inputs to estimate gradient-based feature contributions. For each domain, only attribution vectors for spots assigned to that domain are retained.

Projection and differential aggregation. When the model input consists of principal components, feature-level attribution matrices are projected to gene space using the PCA loading matrix:

$$\boldsymbol{\mathcal{W}}^{M,c}_{\text{gene}} = \boldsymbol{\mathcal{W}}^{M,c}_{\text{PC}} \mathbf{L}, \tag{S9}$$

where $M \in \{\text{IG}, \text{OC}, \text{GS}\}$, $\mathbf{L} \in \mathbb{R}^{m \times G}$, and G is the number of genes represented in the PCA input. This expression applies to PCA without whitening. The implementation additionally rescales component attributions by the square roots of their explained variances when whitening is enabled. For MERFISH, no PCA projection is required.

Rows of $\boldsymbol{\mathcal{W}}^{M,c}_{\text{gene}}$ contain the gene-level attributions for spots assigned to domain c. Let $\boldsymbol{\mathcal{W}}^{M}_{i} \in \mathbb{R}^{G}$ denote the gene-level attribution vector of spot i under method M, with gene index j. These vectors are normalized separately for each spot:

$$\tilde{\boldsymbol{\mathcal{W}}}^{M}_{i} = \frac{\boldsymbol{\mathcal{W}}^{M}_{i}}{\|\boldsymbol{\mathcal{W}}^{M}_{i}\|_2 + \epsilon}, \qquad \epsilon = 10^{-8}. \tag{S10}$$

Let $\mathcal{U}_c$ contain spots from the retained neighboring domains with available attributions. The target-domain and pooled-neighbor means are

$$\hat{\mathcal{W}}^M_{c,j} = \frac{1}{|\mathcal{V}_c|} \sum_{i \in \mathcal{V}_c} \tilde{\mathcal{W}}^M_{i,j}, \qquad \hat{\mathcal{W}}^M_{\mathrm{NBR}(c),j} = \frac{1}{|\mathcal{U}_c|} \sum_{i \in \mathcal{U}_c} \tilde{\mathcal{W}}^M_{i,j}. \tag{S11}$$

Each spot's attribution is computed for the logit of its own inferred domain. Consequently, the differential score compares the target domain's contributions with neighboring domains' contributions to their respective logits. If no retained neighbor has available attributions, the implementation pools other available domains; if none is available, the comparison mean is set to zero. The differential score and Borda consensus ranking are defined in the main Methods.

Candidate gene selection. The consensus ranking supports either a fixed top-k_{gene} cutoff or an adaptive threshold. For adaptive selection, let $\mathrm{Score}_{c,j}$ denote the Borda score and define $\ell_{c,j} = \log(\mathrm{Score}_{c,j} + 10^{-10})$. Genes are retained when

$$\ell_{c,j} > \mu_c + \lambda_{\mathrm{sel}} \sigma_c, \tag{S12}$$

where μ_c and σ_c are the mean and population standard deviation of log scores across genes, and the default λ_{sel} is 1.5. Selected genes are ordered by their original consensus scores. Alternatively, the top-k_{gene} strategy returns the highest-ranked genes for each domain. These procedures prioritize candidate domain-associated genes and do not provide statistical significance or establish causal regulation.

S6 Evaluation metrics

Agreement with reference annotations. Domain identification is evaluated by comparing inferred domain labels C with reference annotations Y. The adjusted Rand index (ARI) measures pairwise agreement while correcting for chance [18]. Let n_{ij} be the number of spots assigned to reference class i and predicted domain j, with marginal counts $a_i = \sum_j n_{ij}$ and $b_j = \sum_i n_{ij}$. For N evaluated spots,

$$\mathrm{ARI} = \frac{\displaystyle\sum_{ij} \binom{n_{ij}}{2} - \frac{\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}}{\binom{N}{2}}}{\displaystyle\frac{1}{2}\left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2}\right] - \frac{\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}}{\binom{N}{2}}}. \tag{S13}$$

An ARI of 1 indicates identical partitions, whereas values near 0 indicate agreement comparable to chance.

Normalized mutual information (NMI), homogeneity (HOM), and completeness (COM) provide com-

plementary information-theoretic measures [19]:

$$\mathrm{NMI} = \frac{2I(Y;C)}{H(Y) + H(C)}, \tag{S14}$$

$$\mathrm{HOM} = 1 - \frac{H(Y \mid C)}{H(Y)}, \qquad \mathrm{COM} = 1 - \frac{H(C \mid Y)}{H(C)}, \tag{S15}$$

where H denotes entropy and $I(Y;C) = H(Y) - H(Y \mid C)$ is mutual information. Homogeneity assesses whether each predicted domain contains a single reference class, while completeness assesses whether each reference class is assigned to a single predicted domain. The composite score used in this study is

$$\mathrm{ACC} = \frac{\mathrm{NMI} + \mathrm{HOM} + \mathrm{COM}}{3}. \tag{S16}$$

Here, ACC denotes this composite score rather than the fraction of correctly classified spots. Higher ARI and ACC indicate better agreement with the reference annotations.

Spatial coherence. CHAOS measures within-domain spatial dispersion using nearest-neighbor distances. Before its calculation, each spatial coordinate dimension is standardized to zero mean and unit variance over the evaluated spots. Let $\tilde{\mathbf{s}}_i$ be the standardized coordinates of spot i, $\mathcal{V}_k$ the spots in predicted domain k, and $\mathcal{C}_{>2}$ the domains containing more than two spots. The implemented score is

$$\mathrm{CHAOS} = \frac{1}{N} \sum_{k \in \mathcal{C}_{>2}} \sum_{i \in \mathcal{V}_k} \min_{j \in \mathcal{V}_k \setminus \{i\}} \|\tilde{\mathbf{s}}_i - \tilde{\mathbf{s}}_j\|_2. \tag{S17}$$

Domains containing at most two spots do not contribute to the numerator, while N includes all evaluated spots. Lower values indicate shorter within-domain nearest-neighbor distances.

PAS measures the fraction of spots whose predicted label differs from those of a strict majority of their ten nearest spatial neighbors. Neighbors are identified using Euclidean distance in the supplied spatial coordinates, excluding the spot itself. With predicted label c_i and neighbor set $\mathcal{K}_i$,

$$\mathrm{PAS} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\left[\sum_{j \in \mathcal{K}_i} \mathbb{I}(c_i \neq c_j) \geq 6 \right], \tag{S18}$$

where $\mathbb{I}$ is the indicator function. The combined spatial discreteness score is

$$\mathrm{DIS} = \frac{\mathrm{CHAOS} + \mathrm{PAS}}{2}. \tag{S19}$$

Lower PAS and DIS indicate greater spatial coherence [19].

Batch mixing and domain preservation. The local inverse Simpson's index (LISI) measures label diversity within neighborhoods of the learned latent space [20]. Batch-LISI increases as batches mix, whereas domain-LISI increases as domain labels mix. For spot i, let $L_{b,i}$ and $L_{d,i}$ denote the two

indices. Their normalized values are

$$\tilde{L}_{b,i} = \frac{L_{b,i} - 1}{B - 1}, \qquad \tilde{L}_{d,i} = \frac{L_{d,i} - 1}{D - 1}, \tag{S20}$$

where B and D are the numbers of batches and domain-label categories, respectively, with $B, D > 1$. The domain-preservation term is $1 - \tilde{L}_{d,i}$. We combine it with batch mixing through their harmonic mean:

$$\text{F1LISI}_i = \frac{2\tilde{L}_{b,i}(1 - \tilde{L}_{d,i})}{\tilde{L}_{b,i} + 1 - \tilde{L}_{d,i}}. \tag{S21}$$

Higher F1LISI indicates a better balance between batch mixing and domain preservation. In the DLPFC evaluation workflow, reference cortical-layer annotations are used to calculate domain-LISI. The composite score is denoted F1LISI throughout the manuscript to distinguish it from its individual LISI components.

S7 Baseline methods

We compared our method with several state-of-the-art approaches (Supplementary Table S2). Unless otherwise specified, all baseline methods are run using their official implementations and the default parameter settings recommended by the original authors. Below we briefly summarize each method and highlight dataset-specific parameter choices:

- **STAGATE** [21] is an adaptive graph attention auto-encoder for spatial transcriptomics, which builds a spatial neighbor graph from spot coordinates and learns low-dimensional embeddings by jointly modeling gene expression and spatial context. In our experiments, we dynamically tune the `rad_cutoff` parameter on each dataset so that the resulting graph achieves an appropriate number of neighbors per spot.

- **STAligner** [22] extends STAGATE to integrate multiple spatial transcriptomics slices by aligning their latent embeddings with mutual nearest neighbors (MNNs). It first uses STAGATE as the backbone to obtain spatial embeddings for each slice, then constructs MNN pairs across slices and optimizes a triplet-based objective to pull matched spots together while pushing non-matching spots apart. We tune the `rad_cutoff` parameter in the same manner as in STAGATE to search for an appropriate neighbor range.

- **SEDR** [7] is an unsupervised framework that learns joint embeddings of gene expression and spatial neighborhoods and then applies Harmony-based batch correction in the latent space to integrate multiple spatial transcriptomics slices.

- **STG3Net** [23] introduces a plug-and-play Global Nearest Neighbor (G2N) module to select representative anchor pairs across slices, and combines masked graph convolutional autoencoders with

adversarial learning to perform batch correction while preserving biologically meaningful spatial domains and slice connectivity.

- **DeepST** [24] extracts image features from tissue histology, augments gene expression with spatial and morphological information, and then learns latent representations via a combination of graph neural networks and denoising autoencoders. Domain-adversarial training is further used to reduce batch effects across slices or platforms. DeepST was evaluated on the DLPFC, sagittal mouse brain, and MERFISH hypothalamus datasets.

- **STitch3D** [25] jointly models multiple 2D tissue slices together with a matched single-cell RNA-seq atlas. By learning a shared latent representation on a 3D neighbor graph of spots and reference cell types, STitch3D delineates three-dimensional spatial domains with coordinated gene expression. We provide a suitable single-cell reference where available.

- **CellCharter** [26] summarizes the local cellular neighborhood of each spot into low-dimensional niche signatures and performs probabilistic clustering and downstream statistical testing to detect recurrent niches across samples, technologies and conditions, and to relate niche composition to disease state or clinical outcome.

- **SPIRAL** [27] consists of two consecutive modules: SPIRAL-integration, which uses a graph-based domain adaptation strategy to learn batch-corrected embeddings and denoised gene expression by encoding spatial correlations, and SPIRAL-alignment, which performs cluster-aware optimal transport to align spatial coordinates at the domain level.

- **GraphST** [28] integrates spatial location information and gene expression profiles on a spot-level graph to accomplish spatial clustering, multi-sample integration, and cell-type deconvolution. It employs a graph neural network encoder together with a contrastive objective that pulls embeddings of spatially adjacent spots closer while pushing apart distant or dissimilar spots.

- **MENDER** [29] represents each cell by a multi-range neighborhood context and uses a biology-driven, non-parametric convolution kernel to aggregate conserved spatial patterns across slices and technologies.

- **SpaMask** [30] combines a masked graph autoencoder with masked graph contrastive learning through a shared graph encoder, using node masking to reconstruct gene-expression features from spatial neighbors and edge masking to preserve spatially and transcriptionally coherent relationships, thereby improving spatial domain identification and batch correction across consecutive slices.

- **SpaCross** [31] combines 3D spatial registration with a cross-masked graph autoencoder and a cross-view latent consistency module to reconstruct gene expression while regularizing latent rep-

resentations, and further builds an adaptive hybrid spatial–semantic graph that fuses local spatial neighborhoods with global cluster-level relationships.

- **SpaBatch** [32] combines masked data augmentation with a variational graph autoencoder to learn robust spatially informed embeddings, refines spatial domains via a deep embedded clustering module, and further employs a readout-based triplet learning strategy on mutual nearest neighbors across slices to align batches and preserve conserved tissue structures.

To evaluate the domain-specific gene rankings prioritized by iSpa3D, we compared them with those identified by four representative spatial gene-analysis methods spanning Gaussian process modeling, local spatial autocorrelation, graph deep learning, and interpretable deep learning:

- **SpatialDE** [33] is a statistical framework based on Gaussian process regression that decomposes gene expression variance into a spatial covariance component and an independent noise term. It assesses whether expression variation exhibits spatial dependency via log-likelihood ratio tests against a non-spatial null model. In our experiments, we run SpatialDE with its default settings and rank spatially variable genes (SVGs) according to their statistical significance (q-values).

- **Hotspot** [34] is an unsupervised framework that computes local spatial autocorrelation (local Geary's C) over a spatial k-nearest neighbor (k-NN) graph to uncover spatial expression patterns and group co-expressed genes into modules. We run Hotspot using its default graph-construction parameters and negative binomial/depth-adjusted noise models, prioritizing genes based on their false discovery rate (FDR) and spatial autocorrelation Z-scores.

- **SpaGCN** [35] is a graph convolutional network framework that integrates spatial coordinates, gene expression profiles, and histology images into a weighted graph to identify spatial domains, followed by a spatial-constrained differential expression test to detect domain-specific marker genes. In our evaluation, we follow the author-recommended pipeline to identify spatial domains and use its marker detection module to prioritize domain-enriched genes based on their expression ratios and adjusted p-values.

- **River** [36] is an interpretable deep learning framework designed to prioritize genes exhibiting differential spatial expression patterns (DSEPs). It employs a two-branch neural architecture—separately encoding spatial coordinates and gene expression profiles—coupled with a post hoc attribution mechanism to quantify each gene's contribution to spatial differences. Following the official implementation, we train the predictive model under the default hyperparameter configurations and rank candidate genes based on their post hoc attribution scores.

S8 Implementation details

iSpa3D was implemented in Python using PyTorch 2.5.1 [37] and PyTorch Geometric 2.6.1 with CUDA 12.1. Data preprocessing and analysis used Scanpy 1.10.3 [13], NumPy 1.26.4, Pandas 2.2.3, and scikit-learn 1.5.2. Integrated Gradients, Occlusion, and GradientShap were implemented using Captum 0.8.0. Experiments were performed on a Windows workstation equipped with an Intel Core i7-12700 processor, 64 GB RAM, and an NVIDIA GeForce RTX 3090 GPU with 24 GB memory.

Training configuration. The training implementation uses Adam optimizers for the autoencoder and discriminator, with learning rates and weight decay specified in the experiment configuration. The configuration also controls the number of training epochs, discriminator and autoencoder updates per epoch, gradient clipping, and random seed. Triplet construction and cluster-level contrastive learning are introduced after a warm-up period, and cross-section correspondences are refreshed at the configured interval. The triplet margin is 1.0. Spatial-neighborhood sizes, masking rates, temperatures, and loss weights are specified separately for each dataset.

Reproducibility settings. The training routine initializes the Python, NumPy, and PyTorch random-number generators from the configured seed and enables deterministic PyTorch operations. Separate random states are used for PCA, clustering, and registration. Software and hardware specifications are reported here to support reproduction of the computational environment.

S9 Additional results on DLPFC

Comparisons of the latent representations showed that iSpa3D recovered a continuous, layer-ordered organization, whereas MENDER and DeepST exhibited substantial overlap between cortical layers. GraphST, STAGATE, and SEDR produced more entangled representations in which the expected laminar progression was less distinct (Supplementary Fig. S1). Layer-associated organization and cross-section mixing were also observed in the other donors.

The spatial domain maps across all three donors further supported the correspondence between iSpa3D domains and the reference cortical annotations (Supplementary Fig. S2). SPIRAL produced spatially fragmented domains in some regions, whereas CellCharter merged parts of layer 5 with neighboring layers. iSpa3D retained more coherent laminar boundaries across sections. The domain-associated genes *FABP7*, *MBP*, and *TMSB10* provided additional molecular support for the inferred cortical and white-matter organization.

S10 Additional results across mouse embryonic stages

Across the E9.5, E10.5, and E11.5 embryo sections, several baseline methods recovered broad anatomical regions but showed domain mixing, spatial fragmentation, or mismatches to the reference annotations (Supplementary Fig. S4A). Comparisons of domain accuracy, spatial coherence, and batch integration

are provided in Supplementary Fig. S4B. iSpa3D achieved ARI values of 0.45, 0.34, and 0.48 and ACC values of 0.57, 0.58, and 0.64 at the three stages, respectively.

The joint UMAP representation showed substantial overlap between stages after integration, whereas several competing methods retained stage-separated clusters or fragmented representations (Supplementary Fig. S4C). Spatial expression maps supported the anatomical interpretation of the inferred domains: *Csrp3* marked the cardiac region, *Apoa2* the liver, *Stmn2* neural regions, and *Tnnt1* the dermomyotome (Supplementary Fig. S5). These gene maps complemented the domain-composition and cross-stage correlation analyses described in the main text.

S11 Multi-scale mapping of the mouse brain

Coronal brain organization. We analyzed 35 coronal adult mouse-brain sections spanning the anterior–posterior axis. Registration to the Allen Brain Atlas Common Coordinate Framework provided a shared anatomical reference [38]. iSpa3D achieved the highest overall ARI and ACC among the compared methods and maintained coherent regional assignments across sections (Supplementary Fig. S6A,B; Supplementary Fig. S7).

In the anterior isocortex, iSpa3D distinguished FRP1- and FRP2/3-associated subregions within an area represented by a single coarse isocortex annotation. These subdivisions followed features visible in the histological image (Supplementary Fig. S6C). Along the anterior–posterior axis, hippocampal, olfactory, isocortical, hypothalamic, thalamic, and fiber-tract domains occupied consistent anatomical territories. In posterior sections 032A–035A, the broad retrohippocampal annotation was resolved into entorhinal and presubicular territories (Supplementary Fig. S6D,E).

Domain proportions varied systematically across the 35 sections, reflecting changes in regional composition along the anatomical axis. The genes *Cnih2*, *Lamp5*, and *Tbr1* showed concordant spatial expression patterns in coronal and sagittal sections (Supplementary Fig. S6F,G). Focused analysis of the hippocampal domain showed its appearance, expansion, and subsequent reduction along the coronal series. The domain-associated genes *Prox1* and *Cnih2* remained spatially localized across sections, supporting the continuity of this domain (Supplementary Fig. S8).

Integration of spatially disjoint sagittal sections. We next analyzed anterior and posterior sagittal mouse-brain sections. Only the anterior section had manual annotations for quantitative evaluation; posterior domains were interpreted with reference to the Allen Brain Atlas. iSpa3D and six comparison methods were evaluated using 32 clusters (Supplementary Fig. S9A–C).

CellCharter, SEDR, and STAligner identified broad shared regions but showed fragmentation or poorly resolved boundaries near the interface between sections. iSpa3D preserved an isocortical territory across the interface and resolved subdomains corresponding to layers L1, L2/3, L4, L5, L6a, and L6b. It also retained a distinct diencephalon-associated domain and section-specific regions, including the

anterior main olfactory bulb and posterior cerebellar cortex. The joint UMAP representation showed mixing of spots from both sections, and iSpa3D achieved an ARI of 0.46 on the annotated anterior section (Supplementary Fig. S9D,E).

Representative genes supported these domain assignments. *Gng4* was enriched in the main olfactory bulb, *Ramp3* in the diencephalon-associated domain, and *Vxn* in a continuous isocortical band across the interface. *Pvalb* and *Chn2* showed restricted expression in distinct cerebellar domains (Supplementary Fig. S9F).Additional spatial-domain and embedding comparisons with SpaMask and SpaCross are provided in Supplementary Fig. S3.

Cross-dataset validation. Repeating the sagittal analysis on Section 2 yielded a comparable organization of the anterior and posterior territories (Supplementary Fig. S10). Gene-expression comparisons between Sections 1 and 2 supported reproducible domain assignments. In particular, *S100a5* and *Ttr* localized to the main olfactory bulb glomerular layer and lateral ventricle, respectively, in both datasets [39, 40].

S12 Additional results across hypothalamic sections

The five MERFISH sections were evaluated against annotations of BST, MPA, MPN, PV, PVH, PVT, V3, and Fx. Additional comparisons showed that several methods did not preserve fine regional boundaries (Supplementary Fig. S11). CellCharter merged part of the fornix with the medial preoptic area and assigned part of the annotated medial preoptic area to PVH. Although SEDR achieved a slightly higher F1LISI score, its domains were more fragmented, whereas iSpa3D achieved the highest mean ARI and ACC.

Across the section series, iSpa3D preserved a continuous V3-associated domain, together with contraction of PVH and expansion of PVT. The ventricular genes *Nnat*, *Mlc1*, and *Aqp4* were consistent with ependymal and tanycyte-related expression, whereas *Irs4* and *Esr1* characterized the neighboring PV domain. After registration, *Nnat* and *Cd24a* formed a column-like distribution along the ventricular wall, while *Irs4* and *Pgr* were distributed in surrounding nuclei [9]. Enrichment of fluid transport, intracellular water homeostasis, and cilium-related processes further supported the ventricular interpretation.

S13 Additional results in the mouse olfactory bulb

Cross-platform comparisons showed that SEDR retained partial separation between Stereo-seq and Slide-seqV2 despite Harmony-based correction. STAGATE and STAligner improved platform mixing but produced less distinct boundaries near the accessory olfactory bulb granular layer (AOBgr). CellCharter merged parts of the accessory olfactory bulb and AOBgr with surrounding regions. iSpa3D recovered consistent olfactory nerve, glomerular, and external plexiform layers across platforms (Supplementary Fig. S12A).

The spatial expression maps of *Slc17a7* and *Cdhr1* showed localized expression across the two platforms (Supplementary Fig. S12B). Genes prioritized for the accessory olfactory bulb and mitral cell layer (Clusters 3 and 9) were associated with GO biological processes including axon development, GABAergic synaptic transmission, and neuron projection morphogenesis (Supplementary Fig. S12C).

S14 Additional analyses of Alzheimer's disease and axolotl development

Disease-associated hippocampal domains. In STARmap PLUS sections from AD and control mice, iSpa3D recovered major hippocampal compartments across ages and disease conditions. *Plp1* and *Bin1* were associated with the oligodendrocyte-rich domain, whereas *C1ql3* and *Cplx2* were associated with the dentate gyrus. Subclustering of the microglia-associated domain identified a population, Micro3, that expanded in AD and expressed *Trem2*, *Apoe*, and *Cst7* (Supplementary Fig. S13). These observations relate the inferred domains to disease-associated differences in cell populations and gene expression.

Axolotl telencephalon development. Additional comparisons evaluated iSpa3D against CellCharter, MENDER, SEDR, and STAligner in adult and juvenile axolotl sections. iSpa3D achieved ARI values of 0.537 and 0.455 in the adult and juvenile sections, respectively, exceeding the corresponding values of the four comparison methods shown in Supplementary Fig. S14A. Spatial maps highlight Clusters 1 and 16, Cluster 12, and Cluster 17 in both sections (Supplementary Fig. S14B).

The gene-expression dot plot shows the mean expression and fraction of expressing cells for *ZIC1*, *TAC1*, *SCGN*, and *AGT* across domains. *TAC1* was enriched in Cluster 12, while *ZIC1* and *SCGN* showed prominent expression in Cluster 17. *AGT* was expressed in several domains, including Clusters 16 and 17 (Supplementary Fig. S14C).

S15 Ablation studies and parameter sensitivity

Contributions of model components. Ablation experiments on the 12 DLPFC sections compared the full model with variants lacking complementary masking, CLCL, or the debatching module. The radar plots show that the effects of component removal vary across sections and metrics, with the full model improving ARI, ACC, or DIS on multiple sections (Supplementary Fig. S15A). These comparisons indicate that the components contribute to model performance, although the full model does not achieve the best value for every section and metric.

Parameter sensitivity and computational requirements. Masking rates from 0.1 to 0.9 were evaluated across the 12 DLPFC sections using ARI, ACC, and DIS (Supplementary Fig. S15B–D). The highest mean ARI among the tested settings occurred at $\rho = 0.3$, with variability across sections shown by the standard-deviation band (Supplementary Fig. S15E). Additional analyses assessed sensitivity to spatial-neighborhood size and PCA dimensionality (Supplementary Fig. S16). Runtime and memory comparisons are reported in Supplementary Table S3.

2 Table

Table S1: Summary of spatial transcriptomics datasets used in this study.

Platform	Tissue	Section	Spots	Genes	Reference
10x Visium	Human dorsolateral prefrontal cortex (DLPFC)	151507	4,226	33,538	[2, 3]
		151508	4,384		
		151509	4,789		
		151510	4,634		
		151669	3,661		
		151670	3,498		
		151671	4,110		
		151672	4,015		
		151673	3,639		
		151674	3,673		
		151675	3,592		
		151676	3,460		
Stereo-seq	Sagittal mouse brain (SMB)	Section 1 anterior	2,695	32,285	[4, 7, 12]
		Section 1 posterior	3,355		
		Section 2 anterior	2,825		
		Section 2 posterior	3,289		
	Mouse embryo (ME)	E9.5_E1S1	5,913	23,015	
		E10.5_E2S1	8,494	22,385	
		E11.5_E1S1	30,124	26,854	
	Axolotl brain (AB)	Control_Juv	11,698	19,081	
		Adult	8,243	22,144	
	Mouse olfactory bulb (MOB)	/	19,109	27,106	
Slide-seqV2	Mouse olfactory bulb (MOB)	/	20,139	21,220	[10]
STARmap PLUS	AD Mouse Brain (ADMB)	8 month control	8,506	2,766	[11]
		8 month disease	8,186		
		13 month control	8,034		
		13 month disease	10,372		
MERFISH	Mouse hypothalamic preoptic (MHP)	Bregma –0.04 mm	5,488	155	[8, 9]
		Bregma –0.09 mm	5,557		
		Bregma –0.14 mm	5,926		
		Bregma –0.19 mm	5,803		
		Bregma –0.24 mm	5,543		
ST	Coronal Mouse Brain (CMB)	01A–35A	152–620	23,371	[6, 41]

Table S2: Overview of comparative spatial domain identification methods.

Method	Methodology	Input Data	Link
STAGATE [21]	Single-slice representation learning	Spatial location data Gene expression data	GitHub
SEDR [7]	Single-slice representation learning	Spatial location data Gene expression data	GitHub
GraphST [28]	Single-slice representation learning	Spatial location data Gene expression data	GitHub
DeepST [24]	Single-slice representation learning	Spatial location data Gene expression data Histology information	GitHub
STAligner [22]	Multi-slice integration and batch correction	Spatial location data Gene expression data	GitHub
SpaBatch [32]	Multi-slice integration and batch correction	Spatial location data Gene expression data	GitHub
STG3Net [23]	Multi-slice integration and batch correction	Spatial location data Gene expression data	GitHub
SPIRAL [27]	Multi-slice integration and batch correction	Spatial location data Gene expression data	GitHub
SpaMask [30]	Single-slice representation learning and 3D tissue modeling	Spatial location data Gene expression data	GitHub
SpaCross [31]	Multi-slice integration and 3D tissue modeling	Spatial location data Gene expression data	GitHub
STitch3D [25]	3D tissue modeling	Spatial location data Gene expression data	GitHub
CellCharter [26]	Microenvironment discovery	Spatial location data Gene expression data	GitHub
MENDER [29]	Microenvironment discovery	Spatial location data Gene expression data	GitHub

Table S3: Computational efficiency and memory consumption across datasets.

Dataset	**Spots**	**MR (s)**	**GMU (MB)**	**MC (MB)**
DLPFC_Donor_1	18,033	205.78	407.76	440.00
DLPFC_Donor_2	15,284	171.91	314.27	356.00
DLPFC_Donor_3	14,364	166.74	332.19	390.00
Mouse embryo	44,531	676.94	1,620.68	1,730.00
Sagittal_Mouse_Brain_Section1	6,050	112.97	715.51	818.00
Sagittal_Mouse_Brain_Section2	6,114	125.33	724.68	816.00
Axolotl brain	19,941	286.91	878.48	916.00
Mouse olfactory bulb	39,248	547.42	984.41	1,038.00
AD Mouse Brain	35,098	456.09	1,026.42	1,066.00
Mouse hypothalamic preoptic	28,317	461.76	607.01	646.00
Coronal Mouse Brain	17,086	308.89	676.69	710.00

MR denotes the model runtime measured as wall-clock training time. GMU represents the peak GPU memory allocated during training. MC indicates the peak GPU memory reserved by the caching allocator.

3 Supplementary Figures

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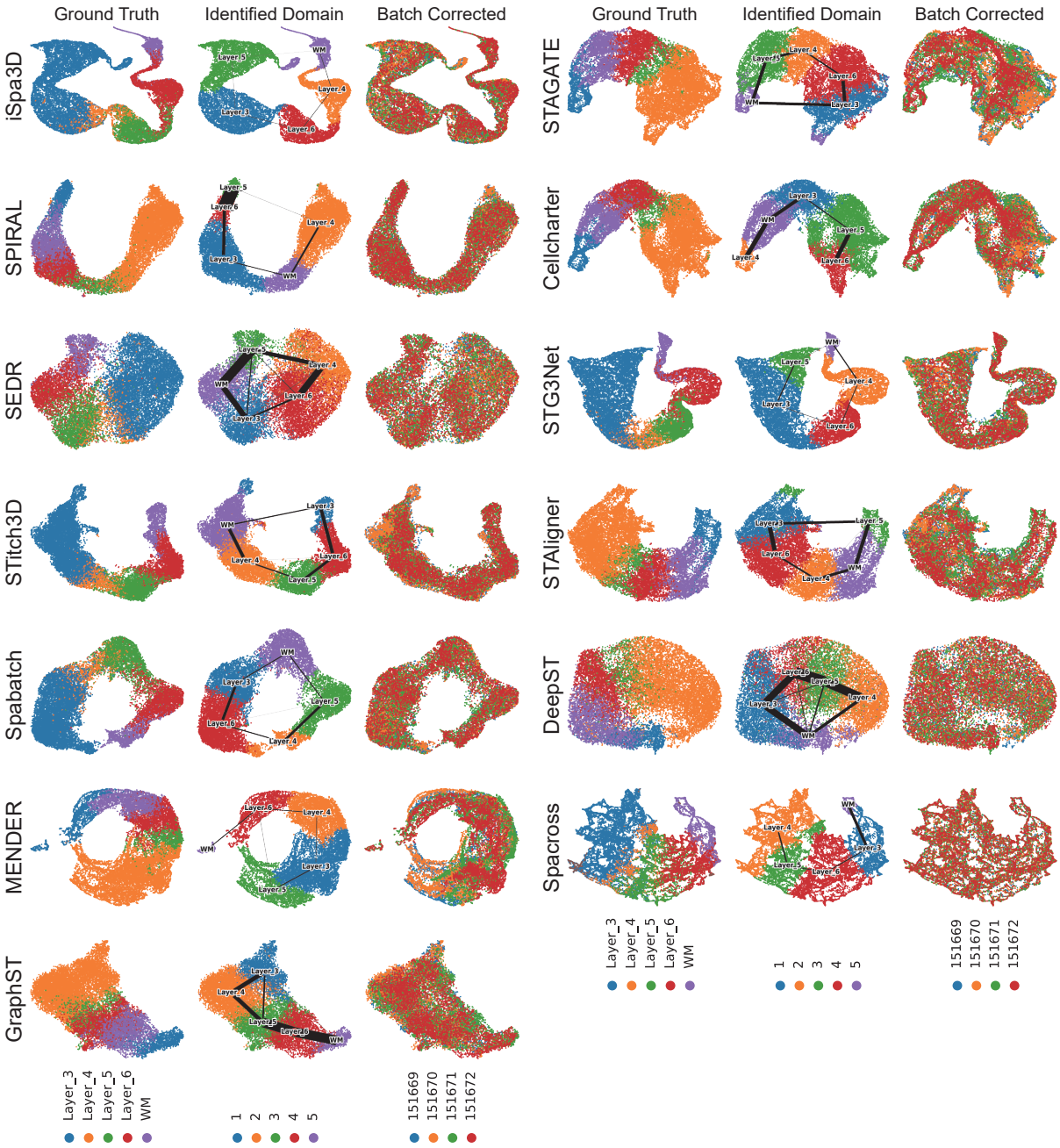

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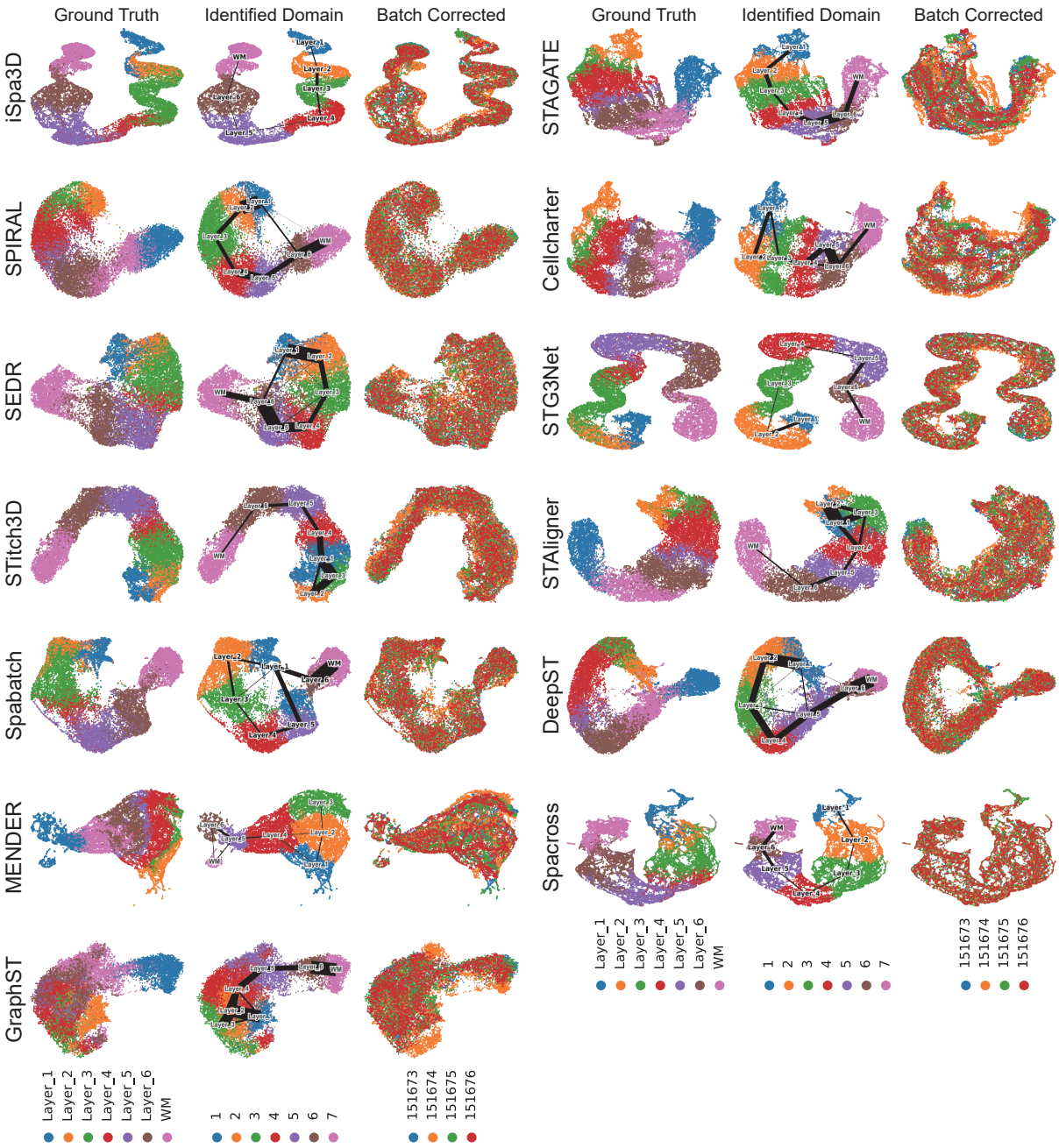
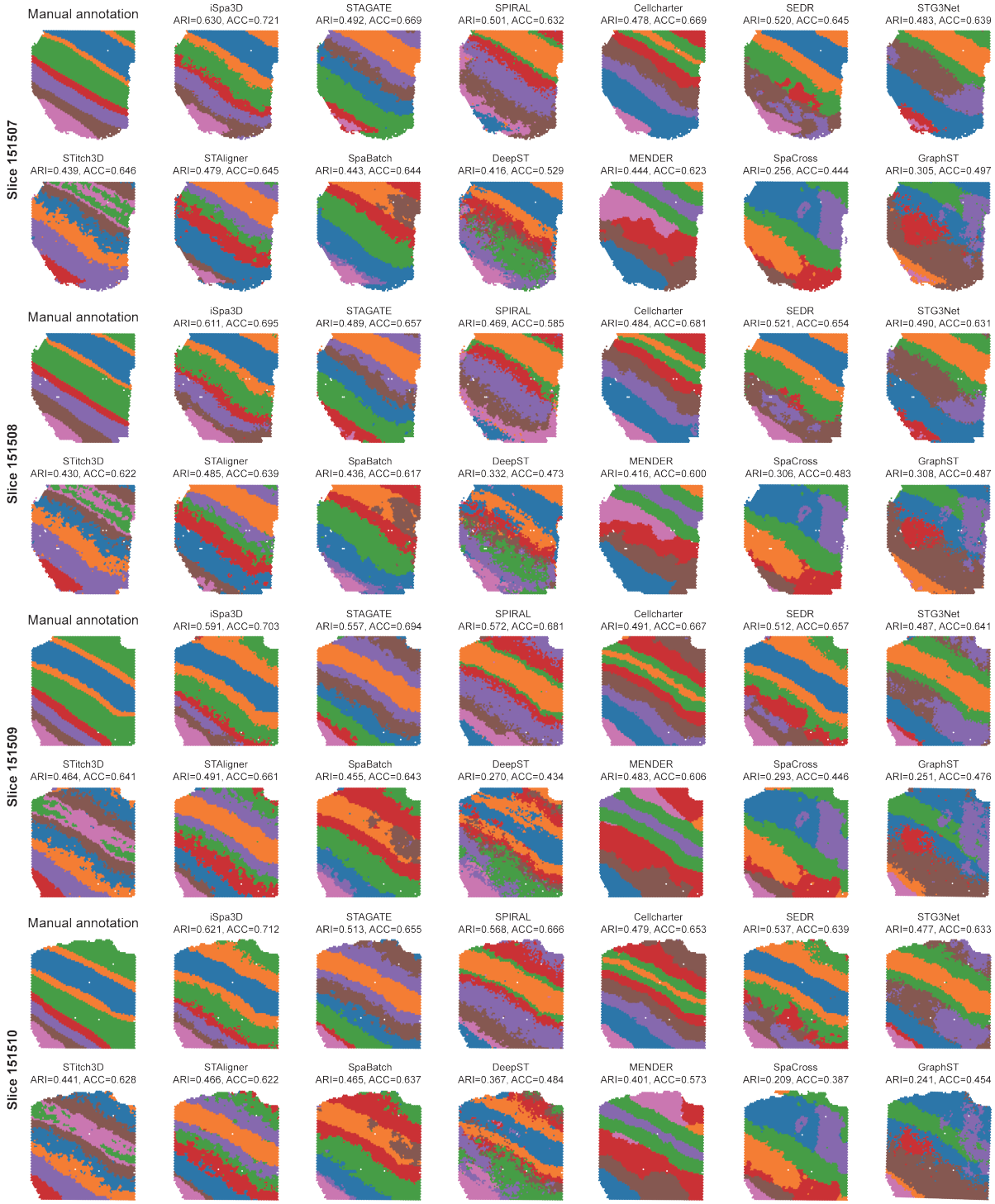


Figure S1: UMAP visualization of the embeddings from the DLPFC dataset, colored by reference annotations (left), spatial domains and PAGA trajectories (middle), and batch-corrected results (right).



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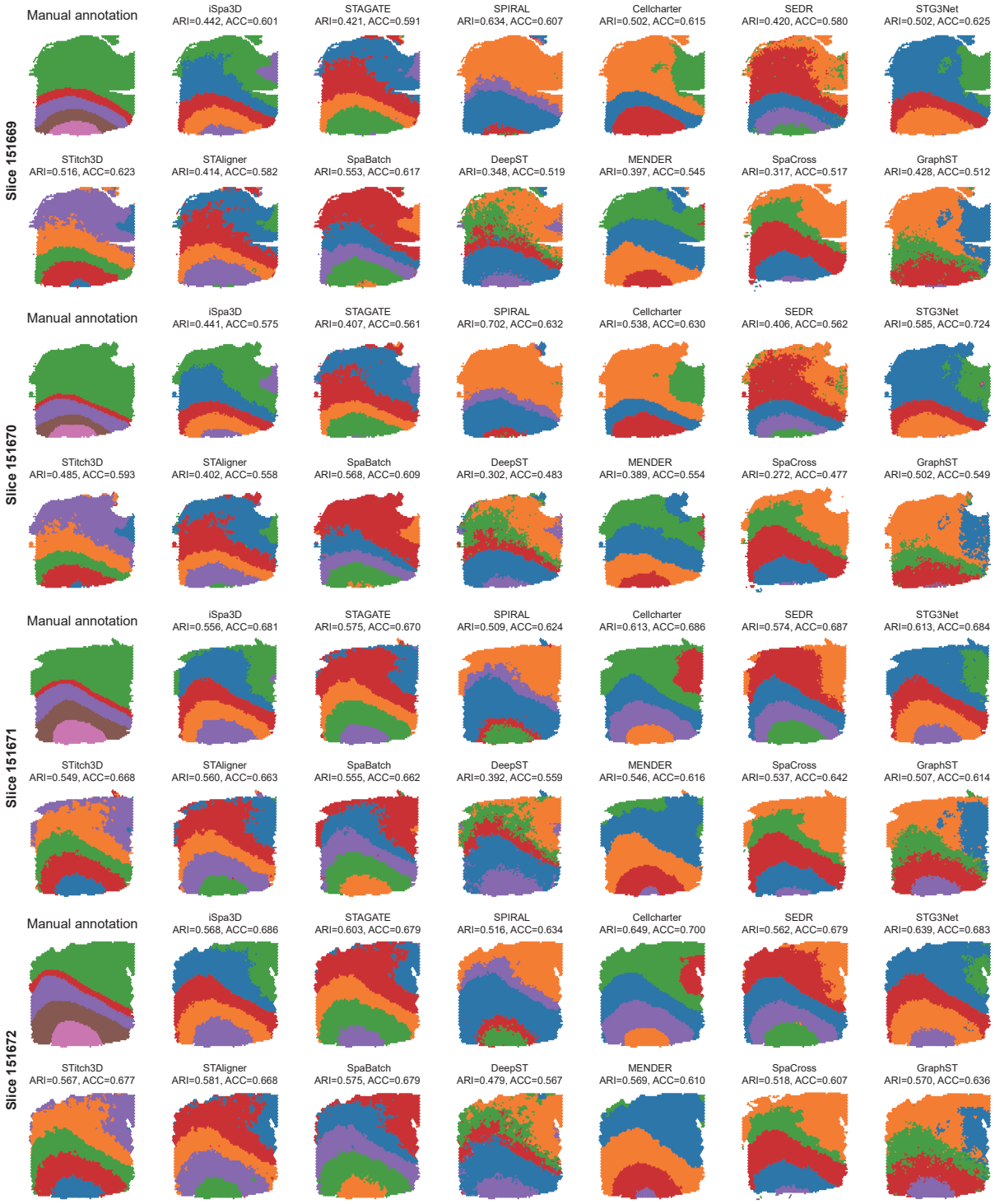

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Figure S2: Comparison of spatial domains by clustering assignments via iSpa3D, various methods, and manual annotation in all 12 sections of the DLPFC dataset.

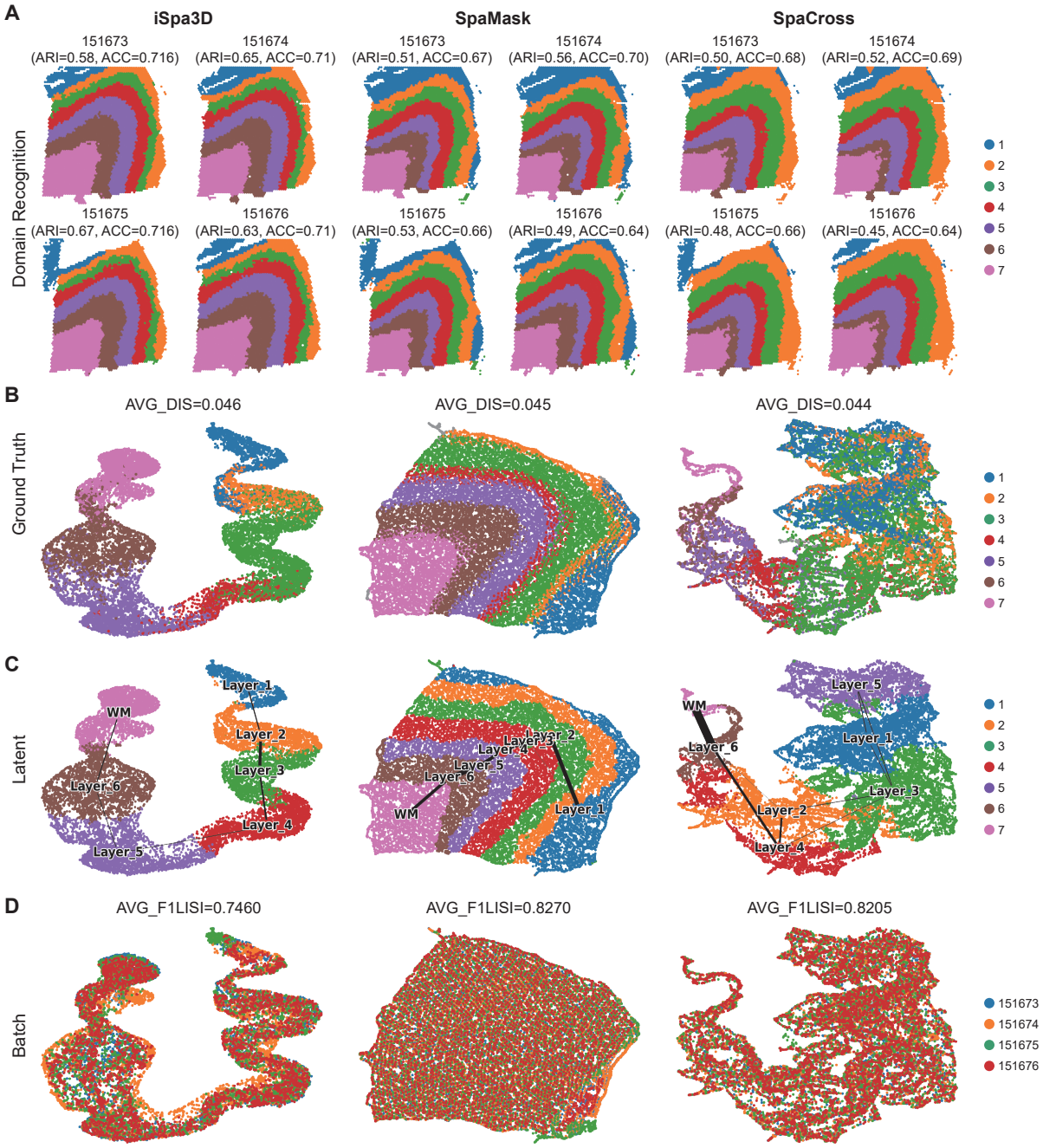

Figure S3: (**A**) Spatial domain prediction results of representative methods (iSpa3D, SpaMask, and SpaCross) across multiple mouse brain sagittal sections. (**B**) Ground truth spatial domain annotations visualized in a unified UMAP space. (**C**) Latent embedding space learned by each method, illustrating the structure and separability of integrated representations across sections. (**D**) Batch mixing visualization showing cells from different sections.

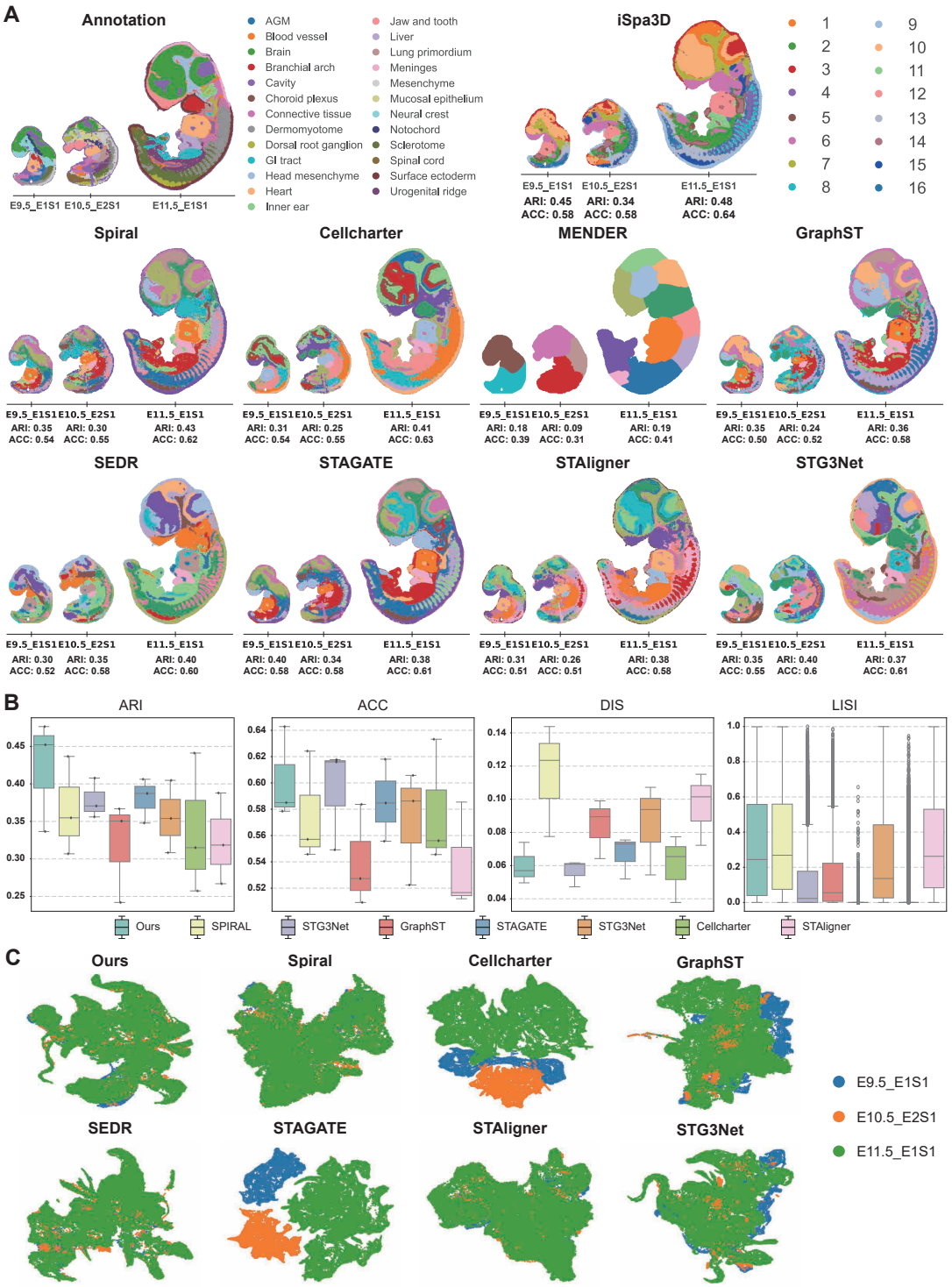

Figure S4: Evaluation of cross-stage spatial domain identification and batch integration on the mouse embryo dataset. (**A**) The top panel displays the ground truth anatomical annotations alongside the spatial domains identified by iSpa3D across three developmental stages (E9.5, E10.5, and E11.5), with corresponding ARI and ACC scores. The lower panels present the spatial clustering results of eight baseline methods for visual benchmarking against the ground truth. (**B**) Boxplots illustrate the statistical comparison of iSpa3D (Ours) against baseline methods across four key metrics. (**C**) Low-dimensional latent representations of spots from all three developmental stages are visualized via UMAP and colored by Stage. This visualization demonstrates the effectiveness of iSpa3D and competing methods in aligning datasets across different temporal batches while preserving biological structures.

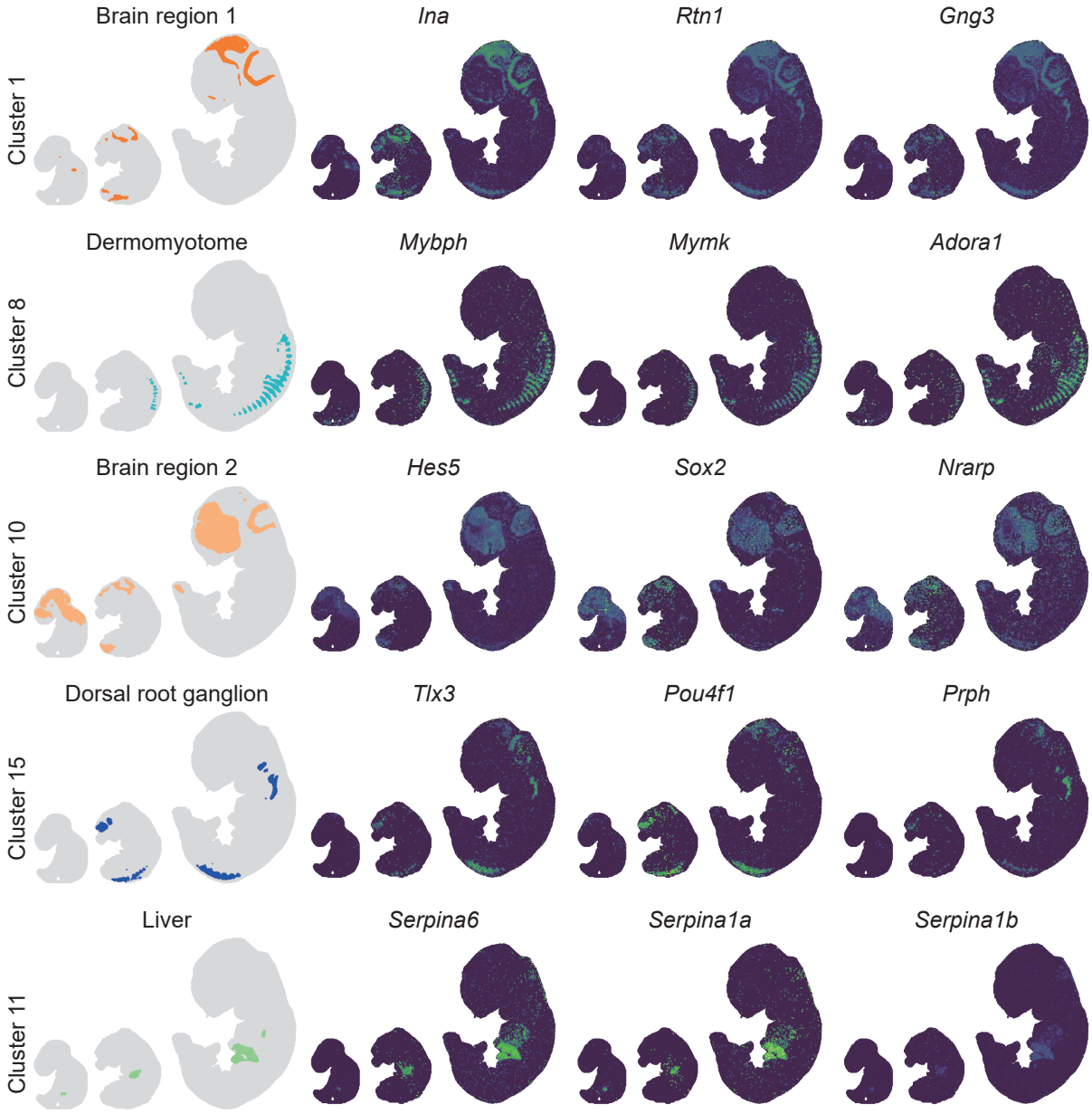

Figure S5: Visualization of domain-specific spatially variable genes (SVGs) identified by the iSpa3D interpretability module in the mouse embryo. Each row corresponds to a distinct spatial domain identified by the model. The leftmost column displays the spatial distribution of the specific cluster across three developmental stages (E9.5, E10.5, and E11.5). The subsequent columns present the spatial expression heatmaps of representative top-ranked marker genes prioritized by the interpretability module for that specific domain.

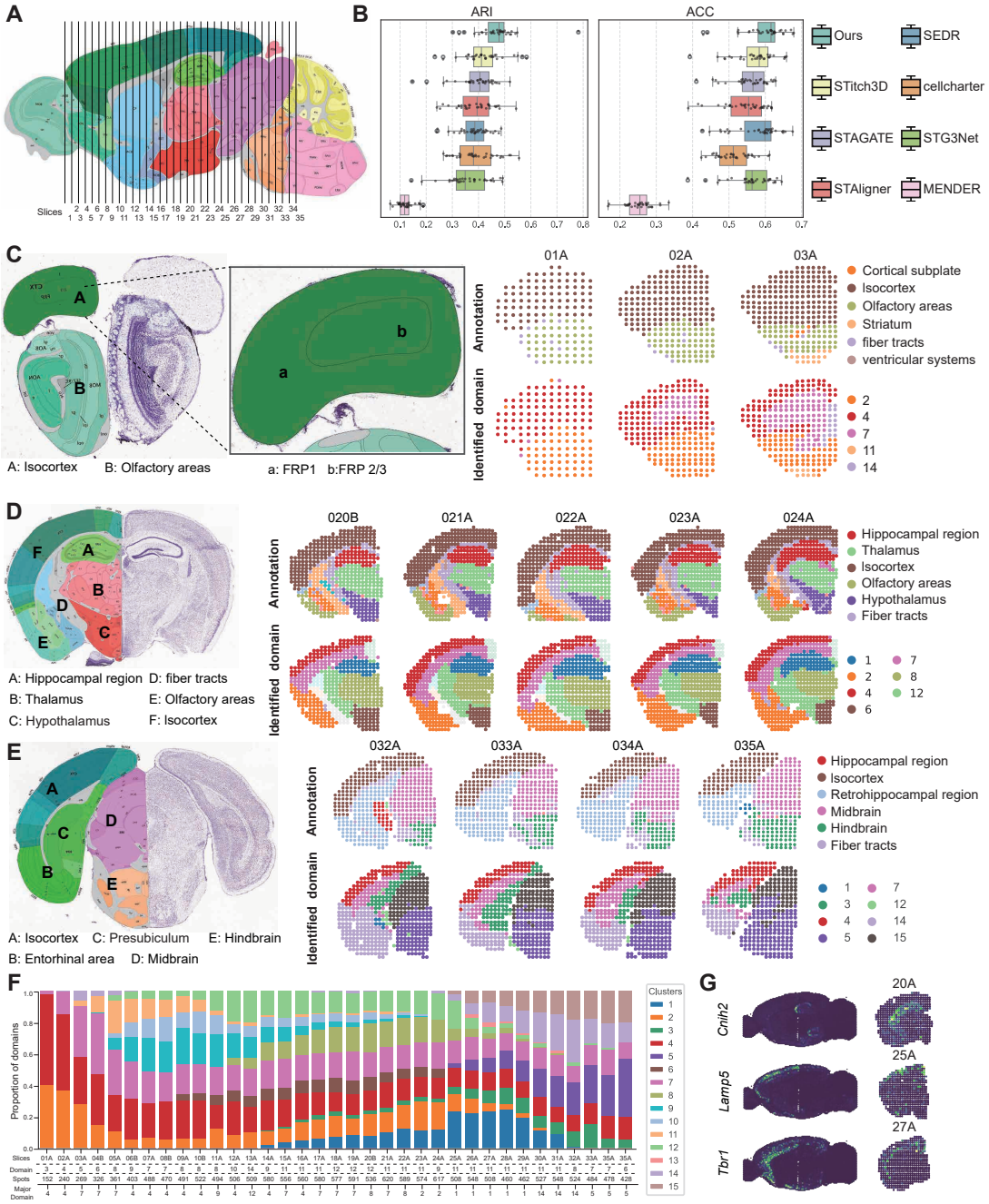

Figure S6: Atlas-guided integration and spatial domain annotation across coronal mouse brain sections. (**A**) Registration of 35 coronal sections to the Allen Mouse Brain Reference Atlas to establish a common anatomical coordinate frame. (**B**) Benchmark performance of ARI and ACC. (**C**) Zoom-in view of the anterior cortex. Atlas labels delineate major territories (e.g., isocortex and olfactory areas), while iSpa3D further resolves finer subregions beyond the coarse atlas boundaries. (**D**) Representative sections along the anterior–posterior axis showing atlas labels (top) and iSpa3D domain maps (bottom), capturing coherent regional transitions across slices. (**E**) Posterior brain examples illustrating complex regional differentiation, including separable entorhinal, presubiculum, and midbrain territories. (**F**) Domain composition across all 35 sections, revealing systematic shifts in regional prevalence along the anatomical axis. (**G**) Representative marker genes (*Cnih2*, *Lamp5*, and *Tbr1*) show consistent spatial patterns across sagittal and coronal mouse brain sections.

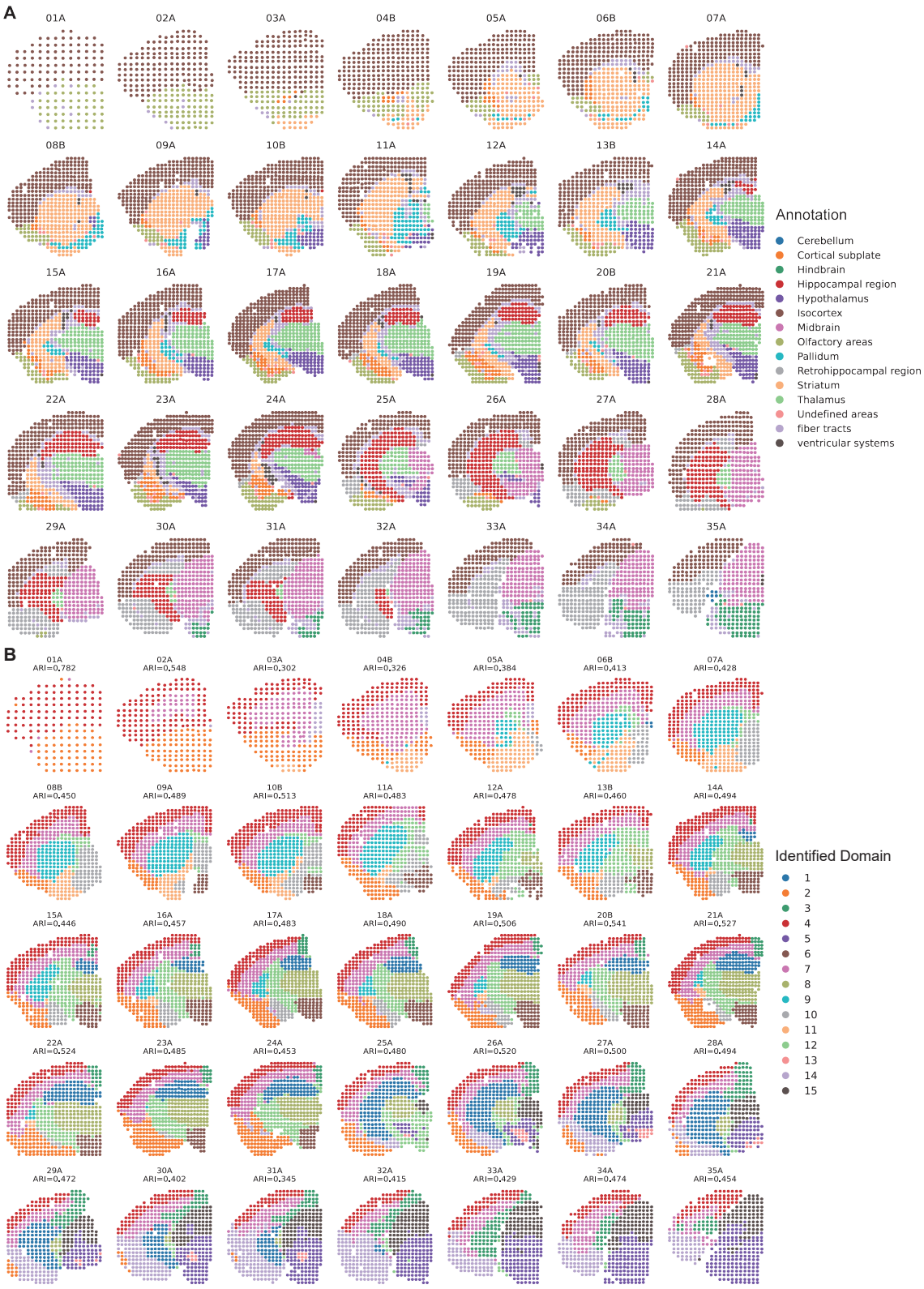

Figure S7: (**A**) Manual annotation of mouse brain data from 35 slices. (**B**) iSpa3D spatial domain identification results.

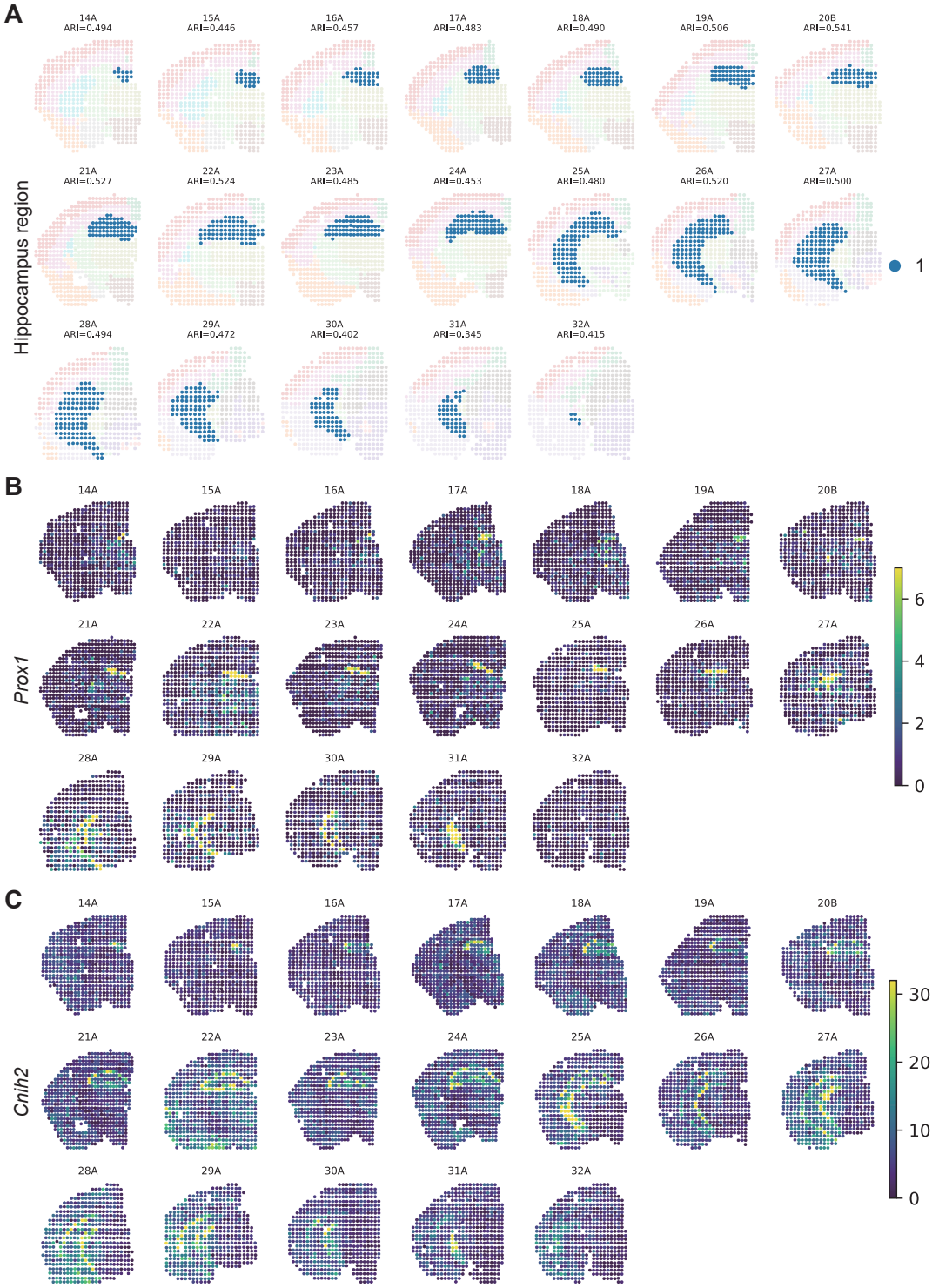

Figure S8: Visualization of the iSpa3D-identified hippocampus region and SVGs detected by the interpretability module in the mouse brain. (**A**) The spatial distribution of the identified hippocampus region (Cluster 1) across continuous coronal sections (from 14A to 32A), spanning from its emergence to disappearance. Blue dots represent spots assigned to the hippocampus domain by iSpa3D. The ARI for each slice is indicated above the corresponding section. (**B**–**C**) Spatial expression heatmaps of representative top-ranked spatially variable genes (SVGs) identified by the interpretability module for the hippocampus domain, visualized using raw gene expression levels. The spatial expression patterns of *Prox1* and *Cnih2* exhibit strong enrichment within the hippocampal region, aligning precisely with the identified spatial domain.

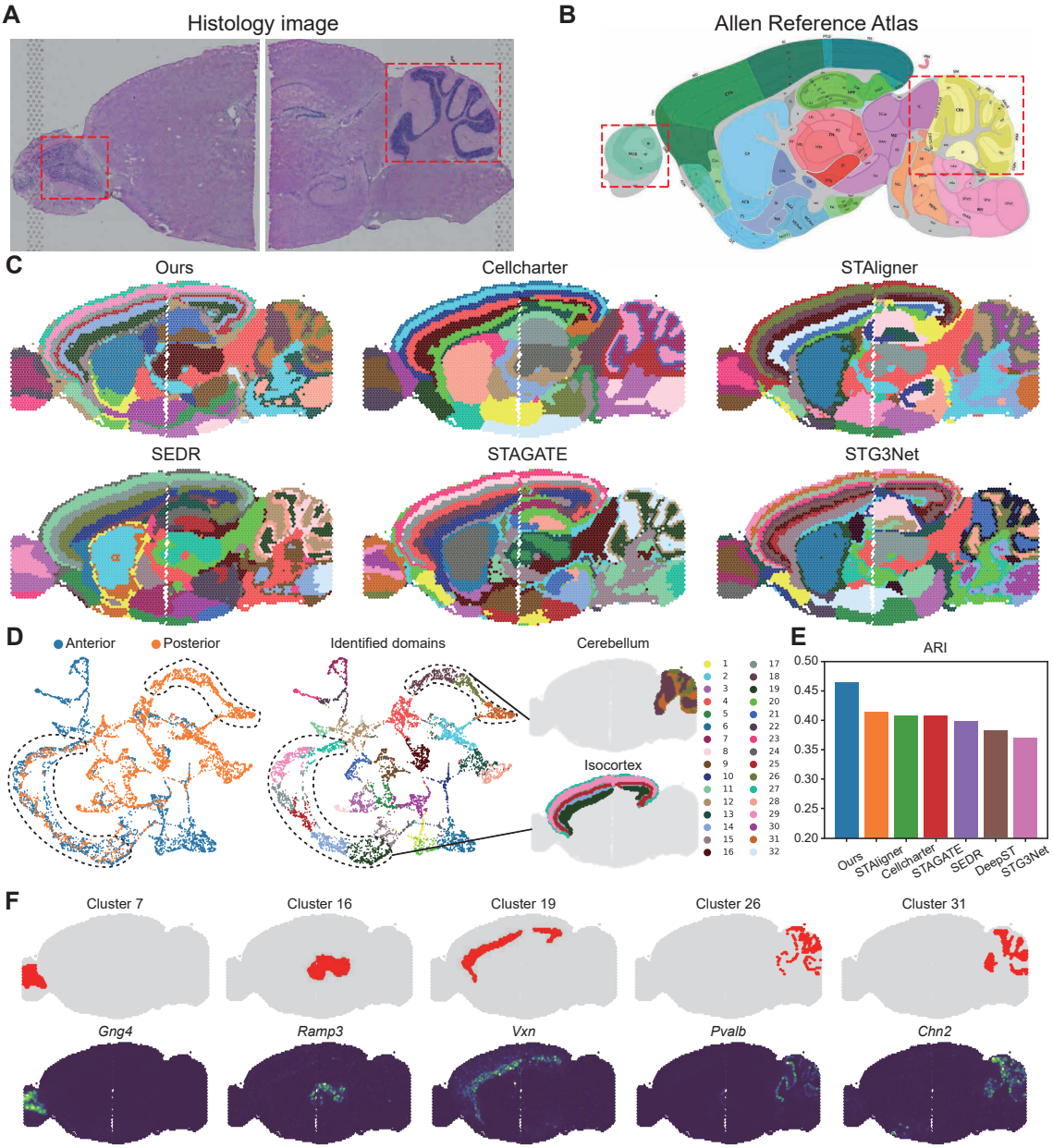

Figure S9: Macro-scale depiction of the sagittal mouse brain and validation of fine subdomains. (**A**) H&E images of the anterior and posterior sections. (**B**) Sagittal anterior and posterior images from the Allen Mouse Brain Atlas. (**C**) Results of spatial domain identification using iSpa3D and other methods on the sagittal anterior and posterior Section 1 of mouse. (**D**) The UMAP represented by spots from iSpa3D is colored by slice index and clustering results. iSpa3D integration adaptively adjusts the shared variation in the whole organization while maintaining the integrity of some unique signals, such as the layered structure of the Isocortex (clusters 14, 17, 19, 25, 27 and 29) and the Cerebellum (clusters 18, 26 and 31). (**E**) Comparison of ARI across methods on the annotated anterior section. (**F**) Representative refined domains and associated genes, including *Gng4*, *Ramp3*, *Vxn*, *Pvalb*, and *Chn2*.

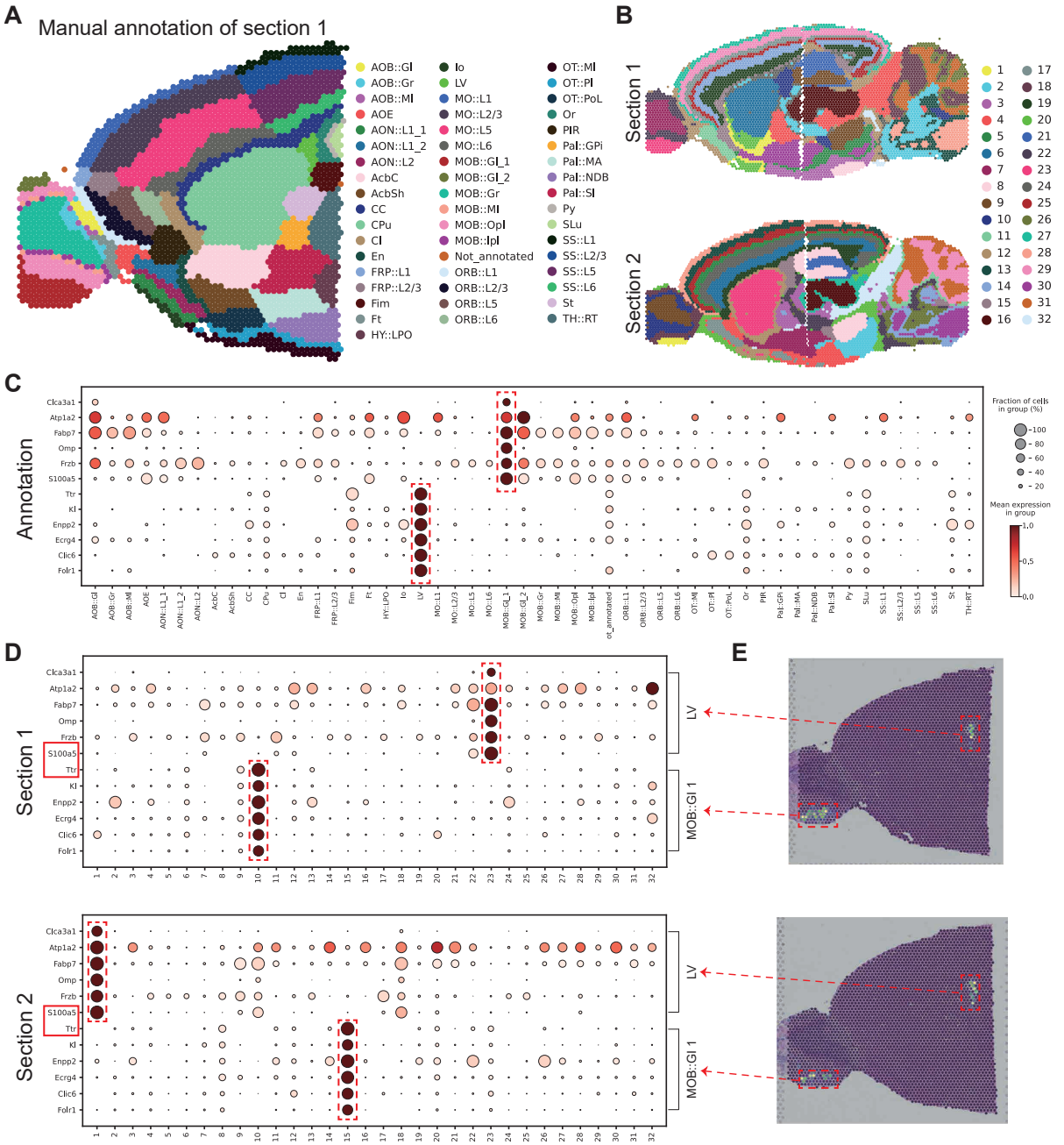

Figure S10: Stitching and fine-grained subdomain identification in mouse brain sagittal sections using iSpa3D. (**A**) Manual annotation of the anterior part of Section 1. (**B**) Integration results of Section 1 and Section 2 using iSpa3D, showing a coherent spatial domain map across the stitched sections. (**C**) Dot plot showing the expression of marker genes across manually annotated anatomical regions, serving as the ground truth reference. (**D**) Dot plots of representative top-ranked spatially variable genes (SVGs) identified by the interpretability module for specific spatial domains in Section 1 and Section 2. The expression patterns in the identified clusters (highlighted in red boxes) align well with the manually annotated spatial domains, specifically the Main Olfactory Bulb Glomerular layer (MOB::Gl) and Lateral Ventricle (LV). (**E**) Visualization of the raw gene expression levels for the marker genes *S100a5* and *Ttr*.

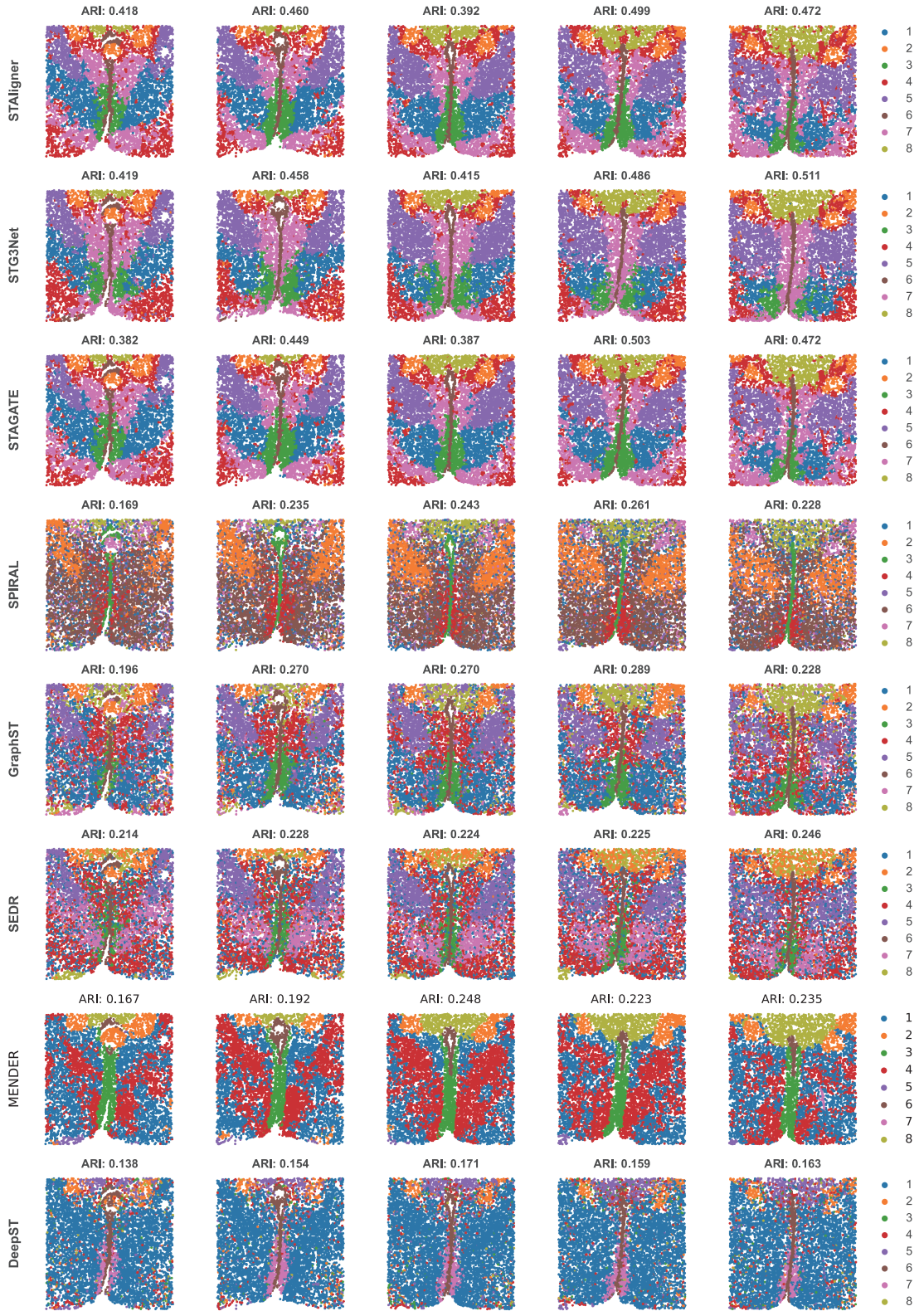

Figure S11: Spatial domains detected by STAligner, STG3Net, STAGATE, SPIRAL, GraphST, SEDR, MENDER and DeepST on the mouse hypothalamus from the MERFISH platform.

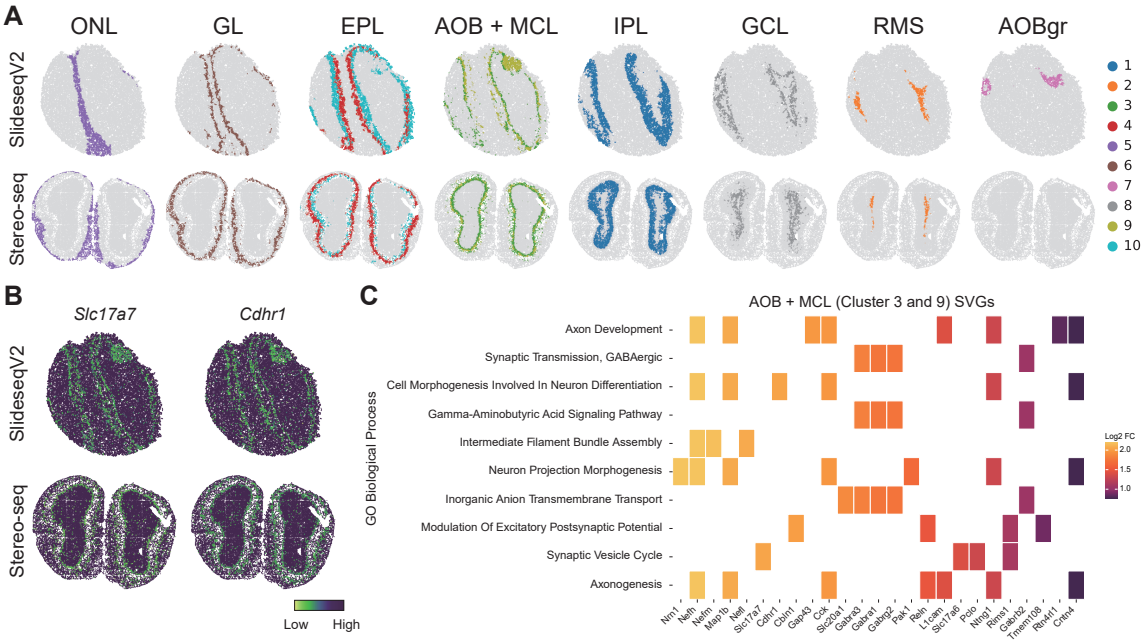

Figure S12: (**A**) Visualization of the spatial distributions of distinct morphological layers identified by iSpa3D across two different platforms: Slide-seqV2 (top row) and Stereo-seq (bottom row). Columns represent specific anatomical regions, including the Olfactory Nerve Layer (ONL), Glomerular Layer (GL), External Plexiform Layer (EPL), Accessory Olfactory Bulb and Mitral Cell Layer (AOB + MCL), Internal Plexiform Layer (IPL), Granule Cell Layer (GCL), Rostral Migratory Stream (RMS), and the fine-grained Accessory Olfactory Bulb granular layer (AOBgr). (**B**) Spatial expression heatmaps of two representative marker genes, *Slc17a7* and *Cdhr1*. (**C**) GO biological processes associated with domain-specific genes prioritized for the AOB and MCL regions (Clusters 3 and 9). Columns represent genes and rows represent GO terms; colored tiles indicate gene–term associations, with color showing $\log_2$ fold change.

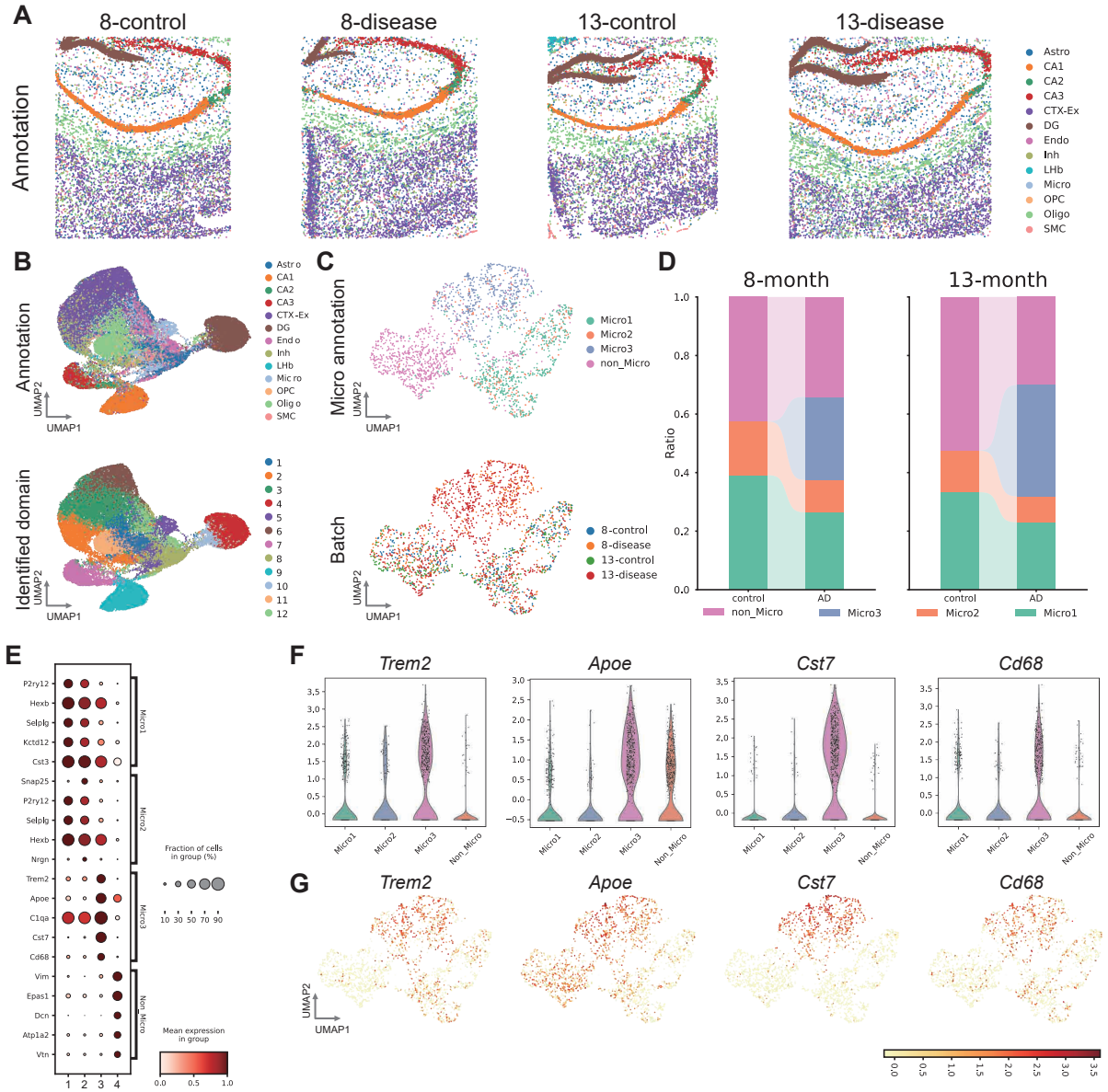

Figure S13: Characterization of disease-associated microglia sub-populations in the Alzheimer's disease (AD) mouse model via iSpa3D domain sub-clustering. (**A**) Spatial distribution of manually annotated cell types across four brain sections representing different disease stages (8 and 13 months) and conditions (Control and Disease). (**B**) UMAP visualization of all single cells colored by manual annotations (top) and iSpa3D-identified spatial domains (bottom), revealing that Domain 5 corresponds to the microglia/immune population. (**C**) Sub-clustering analysis of the cells within iSpa3D-identified Domain 5. The UMAPs display three distinct microglia subtypes (Micro1, Micro2, Micro3) and a non-microglia cluster (top), alongside the batch distribution (bottom). (**D**) Stacked bar plots quantifying the cellular composition of the sub-clusters across different groups. Notably, the Micro3 population exhibits a substantial expansion in the AD (Disease) samples compared to Controls at both time points, suggesting a disease-reactive state. (**E**) Dot plot showing the top differentially expressed genes for each sub-cluster. (**F**) Violin plots and (**G**) feature plots demonstrate that genes such as *Trem2*, *Apoe*, *Cst7*, and *Cd68* are highly enriched in the Micro3 cluster, supporting its interpretation as a pathology-associated microglial population.

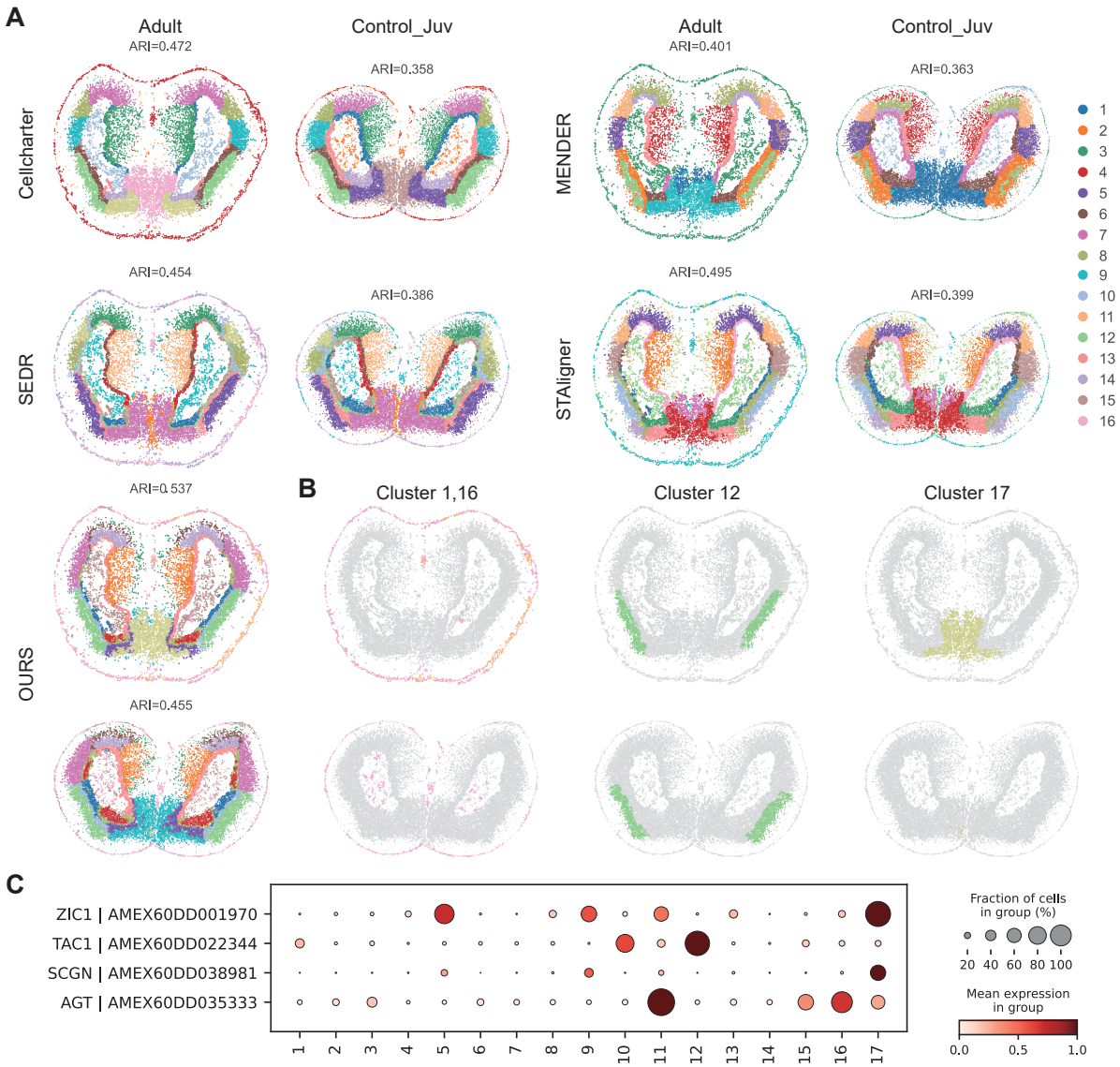

Figure S14: (**A**) Comparison of spatial clustering results between iSpa3D (OURS) and the four methods (CellCharter, MENDER, SEDR, and STAligner) across Adult and Juvenile developmental stages. The ARI is provided for each method. (**B**) Visualization of specific spatial domains identified by iSpa3D (Clusters 1 & 16, Cluster 12, and Cluster 17). (**C**) Dot plot of *ZIC1*, *TAC1*, *SCGN*, and *AGT* across spatial domains. Dot size represents the fraction of expressing cells, and color indicates mean expression. *TAC1* is enriched in Cluster 12, while *ZIC1* and *SCGN* show prominent expression in Cluster 17.

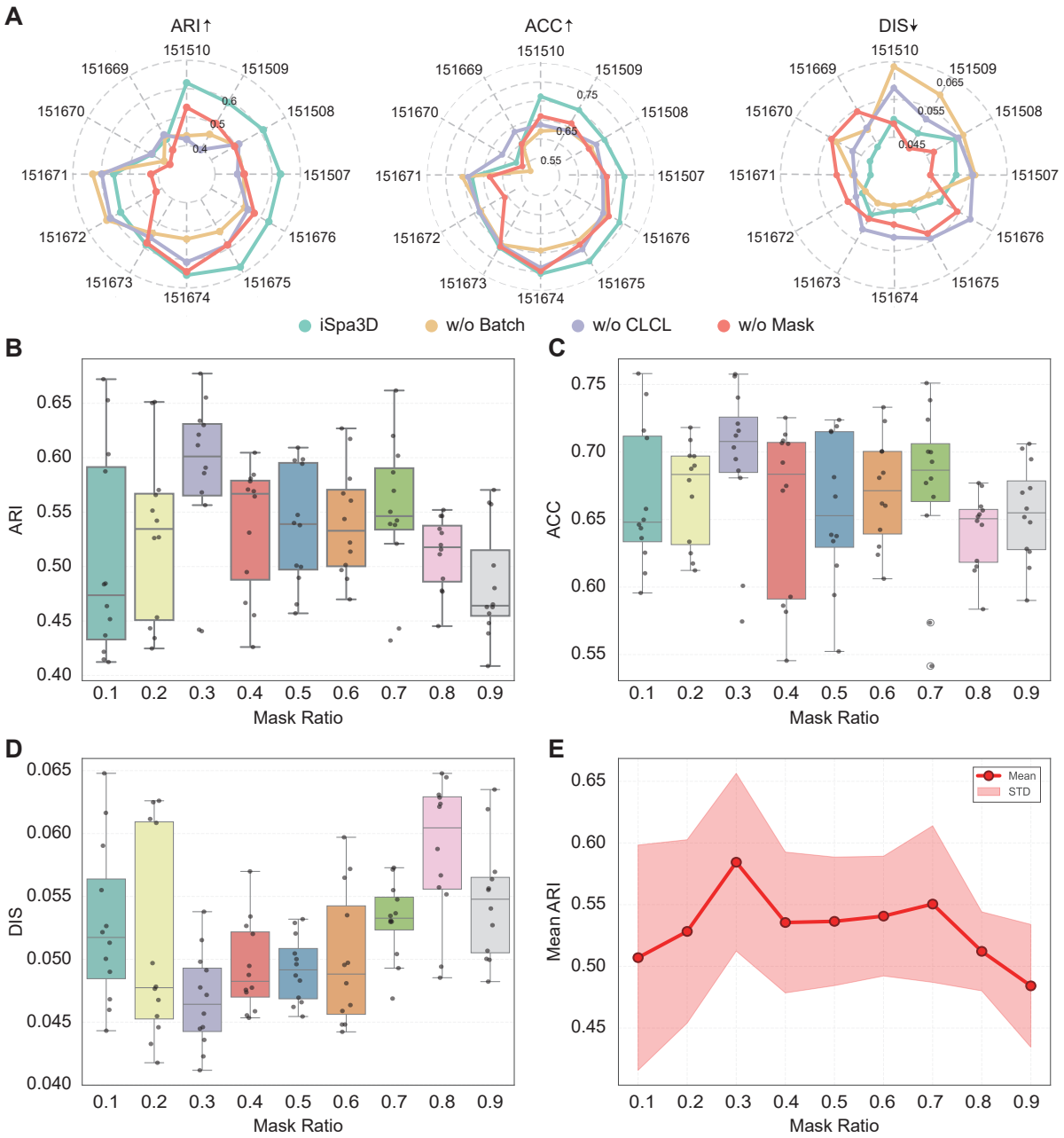

Figure S15: **Ablation studies and hyperparameter sensitivity analysis on the DLPFC dataset.** (**A**) Radar charts comparing the performance of the full iSpa3D model against three ablated variants (w/o Batch, w/o CLCL, w/o Mask) across 12 spatial slices. The axes represent ARI (left, higher is better), ACC (center, higher is better), and DIS (right, lower is better). Each spoke represents a DLPFC section; differences between the full model and ablated variants vary across sections and metrics. (**B**–**D**) Box plots illustrating the impact of the mask ratio (ρ) on model performance metrics: ARI (**B**), ACC (**C**), and DIS (**D**). Experiments were conducted with ratios ranging from 0.1 to 0.9. (**E**) Line plot displaying the mean ARI score with standard deviation (shaded region) across different mask ratios. The highest mean ARI among the tested masking rates is observed at $\rho = 0.3$.

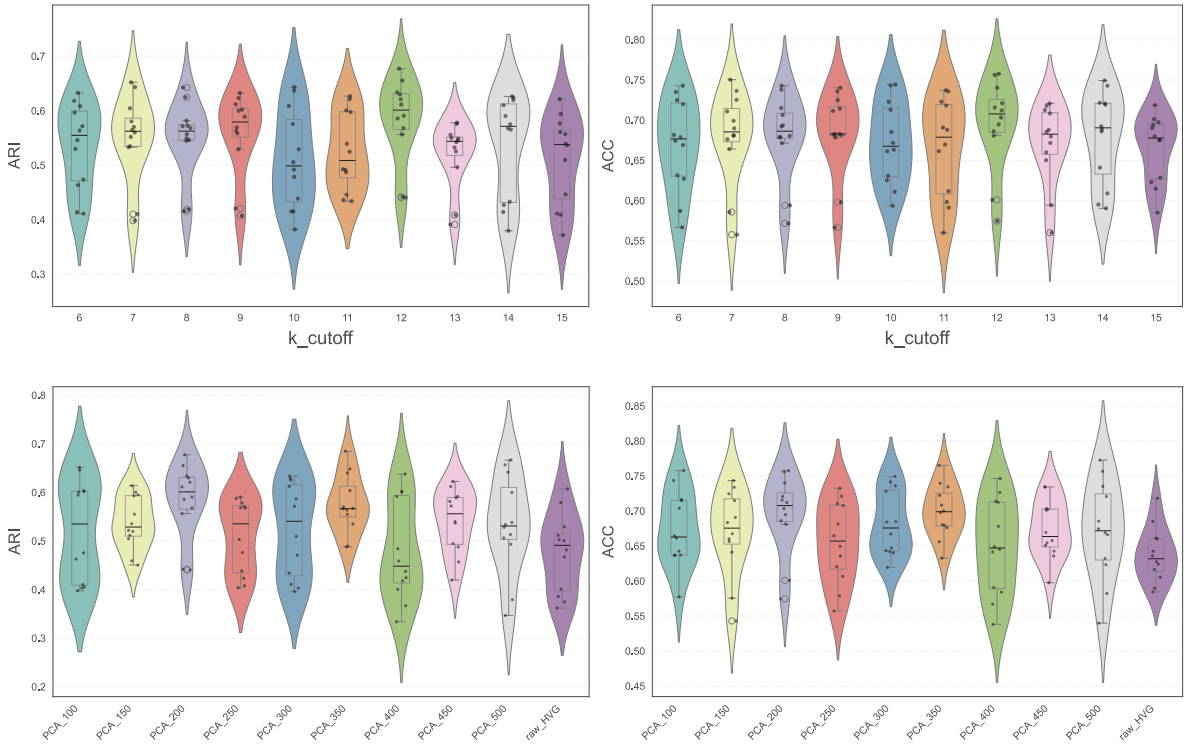

Figure S16: (**Top**) Clustering performance (ARI and ACC) of iSpa3D across 12 DLPFC slices as the number of neighbors in spatial graph construction (`k_cutoff`) varies from 6 to 15. The model exhibits robust performance, with optimal results observed at 12 neighbors. (**Bottom**) ARI and ACC scores under varying PCA dimensionality (ranging from 100 to 500) and a baseline using raw highly variable genes (raw_HVG). The best performance is consistently achieved with 200 PCA components.

4 Supplementary Algorithm

Algorithm S1 iSpa3D: spatial domain identification and 3D reconstruction

Input: Multi-slice expression matrices $\{\mathbf{X}^{(s)}\}$, spot coordinates $\{\mathbf{P}^{(s)}\}$, cluster number K, and training configuration (T_p epochs, T_{warmup} warm-up epochs, refresh interval T_{refresh}, and masking rate ρ).
Output: Spatial domain labels and registered 3D coordinates.

Remove unlabelled spots; preprocess features as in Supplementary Section S2
Obtain $\mathbf{X} \in \mathbb{R}^{N \times m}$; omit PCA for MERFISH
Construct section-wise spatial graphs with edge set $E = \bigcup_s E^{(s)}$ and normalized adjacency $\tilde{\mathbf{A}}$
Initialize encoder F_g, decoder G_d, cluster head, and discriminator D
for $t = 0, \ldots, T_p - 1$ **do**
    Sample complementary sets $\mathcal{V}_m, \mathcal{V}_{cm}$ and construct their masked views
    $\mathbf{Z}_v \leftarrow F_g(\tilde{\mathbf{X}}^{(v)}, \tilde{\mathbf{A}})$ for $v \in \{m, cm\}$
    Independently sample $\mathcal{V}_{\text{rec}}$ and construct $\tilde{\mathbf{X}}_{\text{rec}}$
    $\mathbf{Z}_{\text{rec}} \leftarrow F_g(\tilde{\mathbf{X}}_{\text{rec}}, \tilde{\mathbf{A}})$; $\hat{\mathbf{X}}_{\text{rec}} \leftarrow G_d(\tilde{\mathbf{A}}, \mathbf{Z}_{\text{rec}})$
    Compute $\mathcal{L}_{\text{rec}}$ on $\mathcal{V}_{\text{rec}}$; set $\mathcal{L}_{\text{clu}} = \mathcal{L}_{\text{tri}} = 0$
    if $t \geq T_{\text{warmup}}$ **then**
        Compute $\mathcal{L}_{\text{clu}}$ from cluster contrast and marginal regularization
        if correspondences are uninitialized or due for refresh at interval T_{refresh} **then**
            Rank cross-section candidates by latent dot products and randomly subsample
            Retain same-cluster pairs, remove duplicates, and build the correspondence graph
        end if
        Sample matched positives and same-section negatives to form $\mathcal{T}$
        Compute $\mathcal{L}_{\text{tri}}$ if $\mathcal{T} \neq \emptyset$
    end if
    Update D by minimizing batch cross-entropy $\mathcal{L}_{\text{disc}}$
    $\mathcal{L}_{\text{total}} \leftarrow \lambda_{\text{rec}}\mathcal{L}_{\text{rec}} + \lambda_{\text{tri}}\mathcal{L}_{\text{tri}} + \lambda_{\text{disc}}\mathcal{L}_{\text{disc}} + \lambda_{\text{clu}}\mathcal{L}_{\text{clu}}$
    Update representation-learning parameters, reversing the discriminator gradient
end for
Compute final embeddings $\mathbf{Z}$ and obtain K spatial domains using mclust
Register consecutive sections using latent-space MNNs and SVD-based rigid transformations
Stack registered spot coordinates along the section axis and return coordinates and domain labels

Algorithm S2 iSpa3D: Domain-specific SVG Discovery via Differential Attribution

Input: Gene expression matrix $\mathbf{X}$, spatial graph $\tilde{\mathbf{A}}$, pre-trained encoder F_g, spatial domain labels from Algorithm S1, classifier-training epochs T_f

Output: Domain-specific spatially variable genes (SVGs)

// Supervised Domain Classification
Initialize MLP classifier f_{MLP} for spatial domain prediction
for $t = 0$ to $T_f - 1$ **do**
    Forward pass: $\mathbf{Z} \leftarrow F_g(\mathbf{X}, \tilde{\mathbf{A}})$
    Predict domain labels: $\hat{\mathbf{Y}} \leftarrow f_{\text{MLP}}(\mathbf{Z})$
    Compute cross-entropy loss and update parameters via backpropagation
end for
// Differential Attribution Module
for each spatial domain c **do**
    Define spatial neighborhood $\text{NBR}(c)$ using radius R and a neighbor-ratio threshold τ_{nbr}
    Construct baseline set $\mathcal{B}_c$ from neighbor domain centroids
    for each attribution method $M \in \{\text{IG}, \text{OC}, \text{GS}\}$ **do**
        for each spot i in domain c **do**
            Compute gene-level attribution vector $\boldsymbol{\mathcal{W}}_i^M$ targeting logit $S_c(\mathbf{x}_i)$
            Normalize: $\tilde{\boldsymbol{\mathcal{W}}}_i^M \leftarrow \boldsymbol{\mathcal{W}}_i^M / (\|\boldsymbol{\mathcal{W}}_i^M\|_2 + \epsilon)$, $\epsilon = 10^{-8}$
        end for
        Compute $\hat{\mathcal{W}}_{c,j}^M$ and $\hat{\mathcal{W}}_{\text{NBR}(c),j}^M$ by averaging normalized attributions within the target domain and pooled neighbors
        Compute domain-differential scores: $\Delta_{c,j}^M = \hat{\mathcal{W}}_{c,j}^M - \hat{\mathcal{W}}_{\text{NBR}(c),j}^M$
        Rank genes by differential scores to obtain $r_{c,j}^M$
    end for
    // Consensus Ranking via Borda Count
    Let G be the number of ranked genes; aggregate rankings: $\text{Score}_{c,j} = \sum_M (G + 1 - r_{c,j}^M)$
    Sort genes by $\text{Score}_{c,j}$ in descending order
    Select top-k_{gene} genes or retain genes with $\ell_{c,j} = \log(\text{Score}_{c,j} + 10^{-10}) > \mu_c + \lambda_{\text{sel}}\sigma_c$ (Supplementary Eq. (S12))
end for
return Domain-specific SVGs for each 3D spatial domain

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